Honest Inference for Tensor Contrasts under Tucker and Tensor-Train Rank Bounds

Robustness and adaptation extension: September 11, 2026

Abstract

We study honest confidence intervals for linear functionals of tensor means. Under supplied Tucker or tensor-train rank bounds, the equal-diagonal full-data minimax length is characterized uniformly over positive norm and noise budgets, including arbitrarily weak signals. The corresponding squared-error rate is its square. These upper bounds extend to joint sub-Gaussian errors; directional medians retain the rates over a finite-variance replicate family with a known covariance bound and sufficiently many replicates. A supplied Frobenius approximation allowance adds its size to the interval scale, up to a direct-noise ceiling. For general prescribed loadings we give an orthogonal-surrogate construction with a calibrated residual term, and identify efficiently attained minimax regimes. A growing-rank lower bound and a convex-hull obstruction delimit naive relaxations. At nonzero lower-rank tensors, a shifted mixture illustrates an obstruction from weak additional components. On a different, spectrally separated class, a polynomial SVD/projection procedure selects actual mode ranks without supplied caps and uses an explicit functional-bias certificate. Experiments test its conservatism and failure outside separation, general-loading correction, finite-variance scans, and approximate-rank allowances. Efficient minimax intervals for the original diffuse-loading weak-signal class, sharp geometry for every loading, and general unseparated rank adaptation remain open.

Keywords: honest confidence intervals; linear functionals; Gaussian sequence model; tensor train; Tucker rank; nonconvex parameter space.

1 Introduction

Tensor rank models encode restrictions on several unfoldings of the same parameter. For statistical inference, these restrictions are useful only if their effect on interval length survives uncertainty about the actual ranks. A component can be too weak to identify reliably while still belonging to the class over which coverage is required. Consider a third-order $p \times p \times p$ mean tensor A with $\|A\|_F \leq 1$ and rank at most one. From $n = p$ independent full tensors with standard Gaussian errors, the sample mean has noise level $\tau = p^{-1/2}$. The normalized diagonal sum $p^{-1/2} \sum_i A_{iii}$ has a direct interval of length proportional to τ. Can the simultaneous rank restrictions provide a smaller interval with coverage at every admissible mean, including zero?

Existing theory treats several aspects of this question. Tensor functional inference on regular low-rank strata uses accurate subspace estimation and suitable signal conditions [Xu et al., 2025]. Matrix inference without exact-rank estimation is available under quantitative strength and alignment conditions [Choi et al., 2026]. Classical work establishes nonlinear functional estimation and the limits of honest adaptation on nonconvex parameter sets [Cai and Low, 2004b,a, Collier et al., 2017]. These precedents motivate a narrower comparison: determine whether joint tensor rank constraints

permit a length improvement that cannot be retained by intervals required to cover a larger model defined by a single unfolding.

We study this comparison in the fully observed Gaussian mean experiment, with a known noise variance, a supplied Frobenius bound and supplied upper rank bounds. The loading is a sum of orthogonal product tensors. Honesty means a finite-sample lower bound on coverage over the entire rank-cap class; it includes every lower-rank stratum and every arbitrarily small nonzero singular value. For fixed $K \geq 3$, equal mode size p and $M = \tau\sqrt{p}$, our interval has length $o(\tau)$. By comparison, any honest fixed-radius affine interval on the same class has length at least a constant times τ, and any interval honest over a single-unfolding relaxation has expected length at zero at least a constant times τ. Matricization preserves the data. The latter comparison concerns the parameter space over which coverage is required.

Three mathematical issues govern the argument. Uniform control at rank boundaries requires an envelope that avoids inverses of small singular values. A lower bound for the enlarged model must apply to arbitrary interval procedures, including those using all tensor observations. Finally, optimality depends on which observations a procedure uses: the coordinate diagonal and the full tensor contain different statistical information. This distinction prevents a sharp result for a Gaussian sequence marginal from being mistaken for a sharp full-tensor result.

Canonical Tucker and TT expansions control the sum of squared component weights and hence the $\ell_{2/K}$ size of an orthogonal diagonal. This exponent is familiar from tensor factor geometry [Fan et al., 2023]. A deterministic error envelope for classical soft thresholding [Donoho and Johnstone, 1994] then produces an honest interval. Common-target Gaussian mixtures provide the comparison lower bounds. In the third-order example, the radius is at most $\tau\{8\log(2p/\alpha)/p\}^{1/6}$. At rank one, an exact diagonal-image identity permits a specialization of sparse-sequence theory [Collier et al., 2017]; its lower bound matches this order for diagonal procedures, while full-data analysis requires a different argument. A residual scan exploits off-diagonal information and yields a uniformly short confidence-set projection. A sparse harmonic mixture identifies the larger logarithmic factor in the cubic case. A separate analysis shows that low-degree polynomial estimators need not attain the unrestricted squared-error rate.

The paper develops the following results:

1. An explicit diagonal envelope for Tucker and TT cap classes, and a finite-sample interval valid across their lower-rank strata (Sections 4–5).
2. Two quantitative separations: from fixed-radius affine intervals under the same honesty class, and from all intervals under a single-unfolding relaxation (Section 5).
3. Sharp full-data minimax expected length, both at zero and in the worst case, with rates depending on whether $K = 3$ or $K \geq 4$; the corresponding squared-error rates (Section 5.6).
4. An explicit truncated-scan computation tradeoff and a low-degree polynomial estimation obstruction, distinguished from a lower bound for every efficient confidence interval (Section 5.7).
5. A full norm-budget phase diagram, a residual-fit squared-risk theorem, and loading-dependent bounds with matching flat-support rates (Section 6).
6. Full-data robust-noise and approximate-rank extensions; certified general-loading and efficient regimes; an adaptation obstruction and a separated rank-calibrated procedure with new experiments (Section 7).

2 Related work

Gaussian functionals and approximate sparsity. Optimal recovery and nonconvex functional estimation connect risk to the geometry of the parameter set [Donoho, 1994, Cai and Low, 2004b].

In the Gaussian sequence model $y = \theta + \sigma g$, Collier et al. [2017] give sharp rates for $\sum_i \theta_i$ over ℓ_q balls. In their sparse regime, the squared-error scale is

$$\sigma^2 (r/\sigma)^{2q} \{\log[1 + d(\sigma/r)^{2q}]\}^{1-q}, \qquad \|\theta\|_q \le r, \quad 0 < q \le 1.$$

The specialization $q = 2/K$, $r = M = \tau\sqrt{p}$, $d = p$ and division by p for our normalized contrast yields the squared diagonal-information rate below. We explicitly credit this benchmark and give its confidence-interval derivation. Its transfer to the full tensor experiment would require a different lower bound. Our full-data scan and harmonic mixture provide separate arguments in that experiment.

Honest adaptation and weak ranks. The confidence-interval theory of Cai and Low [2004a] uses between-class moduli and testing to characterize adaptation, while Low [1997] gives foundational limitations. Finite convex-union aggregation is developed by Juditsky and Nemirovski [2020]. Thus, neither nonlinear inference nor failure of uniform adaptation at a model boundary is peculiar to tensors. Matrix confidence sets and detection results further show that attainable rank adaptation depends on the observation design and loss [Carpentier et al., 2018, Carpentier and Nickl, 2015].

In econometric models, Choi et al. [2026] study $Y_{it} = X_{it} M_{it} + E_{it}$ and construct studentized block-average inference with a projection dimension at least the true rank. Their quantitative strength and alignment assumptions remain relevant even though exact rank is not estimated. Armstrong et al. [2026] consider a regression coefficient in a panel with a low-rank interactive effect, using weights and a bias bound for uniform inference without a minimum factor strength. Zhu [2019] establishes weak-factor and rank-adaptation obstructions in a missing-entry experiment. These are different targets and designs from the fully observed tensor contrast considered here.

Regular matrix and tensor functional inference. For tensor regression $Y_i = \langle X_i, A \rangle + \epsilon_i$, Xu et al. [2025] combine a pilot estimator with debiasing and tangent projection to infer general linear functionals. Their regular-stratum results require rank/subspace control and signal/remainder conditions; the principal regression theorem imposes a positive minimum unfolding signal. Its tangent standard deviation is of the form $\sigma \|P_{T_A} L\|_F / \sqrt{n}$. Our replacement is uniform finite-sample coverage on a cap union, at the cost of a narrower loading, known norm/noise bounds and potentially conservative length. No pointwise dominance over that efficient regime is asserted.

Related matrix linear-form inference, geometric inverse-problem inference and limiting distributions under rank caps are developed by Xia and Yuan [2021], Cai et al. [2016], Carpentier and Kim [2018]. Tensor subspace and functional inference includes strong-signal results of Xia et al. [2022], Agterberg and Zhang [2024]; Cai et al. [2023] treats tensor completion and entry/factor uncertainty. Covariance-weighted low-rank regression inference [Ke et al., 2026] and uniform inference for weak matrix signals [Bykhovskaya et al., 2025] provide recent neighboring results. Their observation models and regularity conditions cannot be replaced by the Gaussian location experiment without proof.

Ma et al. [2026] also give a tensor-completion confidence-length lower bound in their Theorem 10. Its class imposes incoherence, a positive minimum unfolding signal, a condition-number cap and nonvanishing tangent-loading alignment. It excludes zero and uses randomly sampled entries; it does not supply a lower bound for the rank-boundary Gaussian location problem here.

Computational restrictions. Low-degree likelihood methods provide restricted computational lower bounds for sparse tensor PCA [Choo and d'Orsi, 2021]. Recent work further studies low-degree estimation thresholds for planted hypergraphs and tensor PCA [Fu and Sohn, 2026]. Our coarse

overlap-moment argument is self-contained and targets the scalar contrast; it does not claim that low-degree polynomials represent every efficient algorithm.

Geometry and finite-sample inference. Orthonormal Tucker and TT representations are standard [De Lathauwer et al., 2000, Oseledets, 2011]. Tensor norms and Euclidean factor penalties already contain the exponent $2/K$ [Friedland and Lim, 2018, Fan et al., 2023]; tangent cones of Tucker and TT varieties have also been studied [Gao et al., 2025, Kutschan, 2018]. We use canonical expansions without estimating tangent spaces. Singular-model likelihood theory [Drton, 2009] and universal inference [Wasserman et al., 2020] establish broader approaches to nonregular validity. Our specific comparison concerns interval length under simultaneous rank constraints, rather than generic existence of a valid interval.

The comparison can be written as

$$\text{regular-stratum functional inference} \quad \longrightarrow \quad \text{cap-uniform inference for a restricted loading,}$$

and, within the present Gaussian experiment,

$$\text{honesty on one unfolding class} \ \longrightarrow \ \text{honesty on the joint tensor class.}$$

The first changes assumptions and loading scope; the second permits a quantitative length separation. The companion survey records 20 closest papers and their mathematical differences, including relevant work available by September 9, 2026.

3 Experiment and parameter classes

Let $K \geq 2$, $p_k \geq 1$, and $D = \prod_{k=1}^{K} p_k$. We observe

$$Z = A + \tau G \in \mathbb{R}^{p_1 \times \cdots \times p_K}, \qquad \mathrm{vec}(G) \sim N(0, I_D), \qquad \tau > 0. \tag{1}$$

The noise level is known in the baseline model. Section 7 states different, explicitly calibrated noise and rank assumptions. Equivalently, Z is the sufficient sample mean of independent full observations $W_j = A + \sigma G_j$, $j = 1, \ldots, n$, with known σ and $\tau = \sigma/\sqrt{n}$. This is an exact Gaussian mean equivalence, not an equivalence to scalar-response tensor regression. The observed variables are the tensors W_j, or Z; A is a fixed unknown parameter. Factor and core representations are nonunique coordinates, not additional latent random variables.

Write $A_{(k)}$ for the mode-k unfolding and $A_{\langle k \rangle}$ for the matrix joining modes $1, \ldots, k$ against $k+1, \ldots, K$. For supplied positive integer caps, define

$$\Theta_{\mathrm{Tu}}(M, \bar{r}) = \{A : \|A\|_F \leq M,\ \mathrm{rank}(A_{(k)}) \leq \bar{r}_k,\ k = 1, \ldots, K\}, \tag{2}$$

$$\Theta_{\mathrm{TT}}(M, \bar{r}) = \{A : \|A\|_F \leq M,\ \mathrm{rank}(A_{\langle k \rangle}) \leq \bar{r}_k,\ k = 1, \ldots, K-1\}. \tag{3}$$

The known bound M is positive. Actual ranks may be smaller, including zero; nonzero singular values may be arbitrarily small. No incoherence or coordinate sparsity condition is imposed. The main experiment assumes Gaussian isotropic noise and exact caps. Appendix C states separately calibrated noise and approximation extensions; missing observations remain untreated.

The prescribed loading has the form

$$L = \sum_{i=1}^{m} w_i V_i, \qquad V_i = \bigotimes_{k=1}^{K} v_{ki}, \qquad v_{ki}^{\top} v_{kj} = \mathbf{1}\{i = j\}, \qquad \|w\|_2 = 1. \tag{4}$$

Here $m \le \min_k p_k$, and each loading family is supplied independently of the data. Thus $\|L\|_F = 1$. Although this is restrictive, L need not be sparse in the observation basis: known orthogonal mode rotations can make it dense. The estimand and relevant observations are

$$\psi(A) = \langle L, A\rangle = \sum_i w_i a_i, \qquad a_i = \langle A, V_i\rangle, \qquad d_i = \langle Z, V_i\rangle = a_i + \tau g_i, \tag{5}$$

where the g_i are independent standard normals.

An interval $C(Z)$ is honest at level $1-\alpha$ on Θ if

$$\inf_{A\in\Theta} \mathbb{P}_A\{\psi(A) \in C(Z)\} \ge 1-\alpha. \tag{6}$$

This is a finite-sample requirement, not a pointwise limit along a fixed-rank stratum. Length and expected length are the primary criteria. For example,

$$\mathcal{L}_0(\Theta) = \inf_{C:\,(6)} \mathbb{E}_0|C|, \qquad \mathcal{L}_{\max}(\Theta) = \inf_{C:\,(6)} \sup_{A\in\Theta} \mathbb{E}_A|C|. \tag{7}$$

The infima range over all measurable intervals, allowing independent randomization. The usual squared Frobenius estimation loss has population risk $R_A(B) = \|B - A\|_F^2 + D\sigma^2$ and empirical risk $R_n(B) = n^{-1}\sum_j \|W_j - B\|_F^2$. We do not minimize these risks or require a full-tensor estimator.

The principal asymptotic comparison in experiment (1) uses $p_k = m = p$, fixed $K \ge 3$, fixed rank caps, and $M = \tau\sqrt{p}$. A concrete sequence is $M = \sigma = 1$, $n = p$, so $D = n^K$. Section 6 removes the relation between M and τ for the sharp full-data results and extends them to flat orthogonal loadings in unequal ambient dimensions.

The structural and probabilistic assumptions have distinct roles, summarized in Table 1. A1–A4 define the inferential problem; A5 is used for the original asymptotic comparisons. The norm/noise relation in A5 is not required in Section 6.

	Assumption	Role and restriction
A1	Full observation with independent Gaussian coordinates and known τ	Gives exact diagonal noise, direct calibration and Gaussian likelihood identities.
A2	Supplied M and Tucker or TT upper rank bounds	Defines the outer honesty class; actual rank and signal strength are unrestricted below it.
A3	Prescribed loading in (4)	Gives orthogonal noise coordinates and the diagonal envelope; data-selected loadings are excluded; arbitrary prescribed loadings are treated separately in Section 7.
A4	Finite-sample uniform coverage in (6)	Coverage includes zero and local sequences approaching lower-rank strata.
A5	Equal dimensions, fixed $K \ge 3$, fixed caps and $M = \tau\sqrt{p}$	Produces the stated vanishing length ratio; not needed for the finite-sample interval.

Table 1: Assumptions, their purpose and the scope they exclude. Known upper bounds are not replaced by same-data estimates anywhere in the stated procedure.

4 Interval construction

Define

$$q = \frac{2}{K}, \quad J_{\mathrm{Tu}} = \min_k \prod_{\ell\neq k} \bar{r}_\ell, \quad J_{\mathrm{TT}} = \prod_{k=1}^{K-1} \bar{r}_k, \quad B = J^{1-q/2} M^q, \tag{8}$$

where J is chosen for the applicable class. Set

$$\lambda = \tau\sqrt{2\log(2m/\alpha)}, \qquad \widehat{\psi}_{\mathrm{NL}} = \sum_i w_i \,\mathrm{soft}_\lambda(d_i), \qquad \mathrm{soft}_\lambda(x) = \mathrm{sign}(x)(|x| - \lambda)_+. \tag{9}$$

The three candidate intervals have radii

$$h_{\mathrm{NL}} = \|w\|_\infty B(2\lambda)^{1-q}, \qquad I_{\mathrm{NL}} = [\widehat{\psi}_{\mathrm{NL}} - h_{\mathrm{NL}}, \widehat{\psi}_{\mathrm{NL}} + h_{\mathrm{NL}}], \tag{10}$$

$$h_G = z_{1-\alpha/2}\tau, \qquad I_G = [\langle L, Z\rangle - h_G, \langle L, Z\rangle + h_G], \tag{11}$$

$$h_0 = M\left(\sum_{i=1}^{s} w_{(i)}^2\right)^{1/2}, \qquad I_0 = [-h_0, h_0], \tag{12}$$

where $|w_{(1)}| \geq \cdots \geq |w_{(m)}|$ and s is the smaller of m and the smallest supplied rank cap. Choose

$$C_* = I_{j_*}, \qquad j_* \in \arg\min_{j \in \{\mathrm{NL}, G, 0\}} h_j, \tag{13}$$

with a fixed rule for ties. This choice depends only on known quantities. Choosing an interval after examining its center or an estimated rank is a different procedure and is not justified by the proof.

The constants in (8) calibrate the threshold procedure (9); the direct alternative (11) supplies a valid fallback. Only m contractions are used. For coordinate loadings, computation reads m tensor entries and takes $O(m)$ operations. Obtaining contractions for a dense rotated loading has additional cost, which can be large. The procedure does not estimate a Tucker core, TT cores, factors or actual ranks.

5 Statistical theory

All numbered theorems and propositions are **PROVED**, with complete derivations in the appendix. The result-status table separates the proved statements from open computational and adaptive questions. The results use the experiment and classes of Section 3. Lower-bound constants are stated for $0 < \alpha < 1/4$ when this simplifies positivity.

5.1 Diagonal geometry and honest coverage

Theorem 1 (Orthogonal diagonal compressibility)**.** *For either class in* (2)–(3)*, under loading* (4)*,*

$$\sum_{i=1}^{m} |\langle A, V_i\rangle|^{2/K} \leq J^{1-1/K} M^{2/K}. \tag{14}$$

No minimum singular value is required.

The proof expands a canonical tensor representation into at most J product terms whose squared component weights sum to $\|A\|_F^2$. For TT tensors, these terms need not be orthogonal. A prefix-weight identity establishes the sum exactly; assuming termwise orthogonality would be an incorrect shortcut. Hölder's inequality and Bessel's inequality then give (14). The rank factor may be loose, particularly for growing caps.

For rank cap one the diagonal image is exactly the ball $\{a : \sum_i |a_i|^{2/K} \leq M^{2/K}\}$, as shown in Lemma 2. Thus the diagonal-only statistical experiment is precisely a classical Gaussian sequence problem on an ℓ_q ball. The full tensor experiment contains additional off-diagonal information. This identity permits a sharp benchmark for the restricted observation experiment; it does not remove the additional information in the rest of the tensor.

Theorem 2 (Finite-sample honest interval)**.** *For every $0 < \alpha < 1$, each interval in (10)–(12) is honest on the applicable cap class. Consequently, C_* is honest and has deterministic length*

$$|C_*| = 2\min\{h_{\mathrm{NL}}, h_G, h_0\}. \tag{15}$$

The event $\max_i |\tau g_i| \le \lambda$ has probability at least $1-\alpha$. On this event soft thresholding satisfies

$$|\operatorname{soft}_\lambda(a_i + \tau g_i) - a_i| \le \min(|a_i|, 2\lambda) \le |a_i|^q (2\lambda)^{1-q}.$$

Summing proves the nonlinear interval bound. No central limit theorem, rank recovery or pilot remainder is needed. Known M is essential to this deterministic calibration.

5.2 Two lower-bound comparisons

For the remainder of this subsection let all dimensions equal p, let $E_i = e_i^{\otimes K}$, and put

$$L = p^{-1/2} \sum_{i=1}^{p} E_i, \qquad z = z_{1-\alpha/2}. \tag{16}$$

Theorem 3 (Fixed-radius affine intervals)**.** *Suppose the honesty class contains 0 and all $\pm M E_i$. If $0 < \alpha < 1/2$ and*

$$C(Z) = [\langle W, Z\rangle + b - h, \langle W, Z\rangle + b + h]$$

is honest, with deterministic W, b, h, then

$$h \ge \frac{(M/\sqrt{p})z\tau}{M/\sqrt{p} + z\tau}. \tag{17}$$

On the rank-one tensor Frobenius ball, the best radius in this affine class is therefore of order $\min(M/\sqrt{p}, \tau)$, up to constants depending on α.

The theorem allows W to use the entire tensor. It does not cover affine centers with data-dependent radii. The matching upper order follows by choosing between I_G and I_0 for rank cap one.

For a nonempty proper subset S of modes, write $A_{[S]}$ for the corresponding bipartition unfolding and define

$$\Theta_S = \{A : \operatorname{rank}(A_{[S]}) \le 1,\ \|A\|_F \le M\}. \tag{18}$$

Theorem 4 (Cost of a single unfolding honesty class)**.** *If $0 < \alpha < 1/4$, every interval honest on Θ_S satisfies*

$$\mathbb{E}_0|C| \ge c_\alpha \min\left\{\frac{M}{\sqrt{p}}, \frac{\tau}{4}\right\}, \qquad c_\alpha = 1 - 2\alpha - \frac{1}{2}\sqrt{\sqrt{8/7} - 1} > 0. \tag{19}$$

The lower order $\min(M/\sqrt{p}, \tau)$ is attained up to α-dependent constants by the direct/structural choice on Θ_S.

The alternatives in the proof are rank one in the chosen matrix unfolding. They generally violate the other tensor rank constraints. This is necessary for the comparison: the theorem measures the cost of requiring coverage on the larger class. For Tucker caps, a mode unfolding is a permissible relaxation; for TT caps, a contiguous cut is permissible. With all tensor rank caps one, every bipartition in (18) contains the original tensor class. Increasing an unfolding cap further preserves the lower bound.

Theorem 5 (Strict length separations). *Let $0 < \alpha < 1/4$, $K \geq 3$ and the rank caps be fixed. Under (16) and $M = \tau\sqrt{p}$, C_* is honest on the Tucker or TT class and*

$$|C_*| \leq 2\tau J^{1-1/K} \left\{ \frac{8\log(2p/\alpha)}{p} \right\}^{(K-2)/(2K)} = o(\tau). \tag{20}$$

In contrast, every fixed-radius affine interval honest on the same class has length at least $2z\tau/(1+z)$. Every interval honest on a class retaining only one constrained unfolding has expected length at zero at least $c_\alpha \tau/4$.

For growing caps, the same conclusion follows if $J^{1-1/K}(\log p/p)^{(K-2)/(2K)} \to 0$. This condition may be restrictive. At $K = 2$ the exponent in (20) is zero, so the argument gives no polynomial separation. Theorem 5 neither asserts that the displayed exponent is optimal nor compares favorably at every strong signal with an efficient oracle projection interval. Figure 1 displays the explicit comparisons and the within-class bracket established next.

5.3 An initial full-data lower bound

Theorem 6 (Within-class lower bound). *Under (1) and (16), for $0 < \alpha < 1/4$, every interval honest on a class containing the rank-one tensor Frobenius ball of radius M satisfies*

$$\mathbb{E}_0|C| \geq (3/4 - 2\alpha)\frac{\min\{M, \tau\sqrt{\log(1+p/4)}\}}{\sqrt{p}}. \tag{21}$$

This is a sparse-coordinate mixture bound. At $M = \tau\sqrt{p}$, for fixed caps and fixed $K \geq 3$, the results imply

$$c\tau\sqrt{\frac{\log p}{p}} \leq \mathcal{L}_0(\Theta) \leq \mathcal{L}_{\max}(\Theta) \leq C\tau\left(\frac{\log p}{p}\right)^{(K-2)/(2K)} \tag{22}$$

for sufficiently large p, with constants depending on K, the caps and α. The ratio of these initial scales grows as $(p/\log p)^{1/K}$. Section 5.6 closes the gaps: the two criteria in (7) have the same rate, with an additional logarithmic factor in the cubic case.

5.4 Optimality when only the diagonal is observed

Let Θ_1 denote the Frobenius ball of tensors with rank at most one, including zero. Restrict the interval to be measurable in $d = (Z_{i,\ldots,i})_{i=1}^p$ and an independent random seed. Write $\mathcal{C}_{\alpha,d}(\Theta_1)$ for such intervals that are honest on Θ_1, and define

$$\mathcal{L}_{0,d}(\Theta_1) = \inf_{C \in \mathcal{C}_{\alpha,d}(\Theta_1)} \mathbb{E}_0|C|, \qquad \mathcal{L}_{\max,d}(\Theta_1) = \inf_{C \in \mathcal{C}_{\alpha,d}(\Theta_1)} \sup_{A \in \Theta_1} \mathbb{E}_A|C|.$$

The next result specializes classical sparse-sequence arguments [Cai and Low, 2004b, Collier et al., 2017] to honest intervals. It is a benchmark for the proposed use of diagonal information, not a new Gaussian-sequence minimax principle.

Proposition 1 (Classical diagonal-information benchmark). *Assume (1), (16) and $0 < \alpha < 1/4$. For each integer $1 \leq s \leq p$, put*

$$b_s = \min\left\{ Ms^{-K/2},\ \tau\sqrt{\log\left(1 + \frac{p}{4s^2}\right)} \right\}, \qquad c_{\alpha,d} = 1 - 2\alpha - \frac{1}{2}\sqrt{e^{1/4} - 1} > 0.$$

Every $C \in \mathcal{C}_{\alpha,d}(\Theta_1)$ *satisfies*

$$\mathbb{E}_0|C| \geq c_{\alpha,d}\frac{sb_s}{\sqrt{p}}. \tag{23}$$

For fixed $K \geq 3$ *and* $M = \tau\sqrt{p}$,

$$\mathcal{L}_{0,d}(\Theta_1) \asymp \mathcal{L}_{\max,d}(\Theta_1) \asymp \tau\left(\frac{\log p}{p}\right)^{(K-2)/(2K)}. \tag{24}$$

The comparison constants depend on K *and* α.

The rate in (24) follows because the effective support size $s \asymp (p/\log p)^{1/K}$ makes (23) match the threshold interval's upper order. The same asymptotic conclusion holds for any fixed Tucker or TT caps: the class contains Θ_1 and Theorem 2 changes the upper bound only by a fixed rank factor. Thus further improvement in the power of p cannot come from recalibrating a procedure restricted to d.

This result does not close the full-data gap in (22). For the common-target tensors $A_S = b_s(\sum_{i\in S} e_i)^{\otimes K}$, two support sets with overlap J have diagonal inner product $b_s^2 J$ but full-tensor inner product $b_s^2 J^K$. The likelihood second moment in the full experiment is consequently different. A valid diagonal lower bound cannot be transferred by ignoring the extra observations. Theorem 7 establishes that they do permit shorter expected length at zero.

Table 2 separates the procedure and parameter classes in the comparisons; Figure 1 displays the corresponding bounds.

Comparison	Information and honesty class	Length statement when $M = \tau\sqrt{p}$
Direct Gaussian interval	All means; prescribed L	$2z\tau$, with exact coverage.
Selected interval	Joint cap class; uses d	At most $2\tau J^{1-1/K}\{8\log(2p/\alpha)/p\}^{\beta}$; $\beta = (K-2)/(2K)$.
Any fixed-radius affine interval	Full tensor; same joint class	Length at least $2z\tau/(1+z)$.
Any interval on one unfolding class	Full tensor; enlarged honesty class	Expected length at zero at least $c_\alpha\tau/4$.
Any diagonal-based interval	Only d; rank-one or fixed-cap class	Optimal expected length at zero and in worst case has order $\tau(\log p/p)^{\beta}$.
Any full-data interval	Full tensor; joint class	Optimal expected length both at zero and in the worst case is $r_K(p)$ in (34).

Table 2: Precisely delimited comparisons for fixed $K \geq 3$ and fixed rank caps. Lower bounds for relaxed honesty classes or restricted observations are not lower bounds for unrestricted full-data procedures on the original class.

5.5 Using the full tensor at the zero mean

The diagonal rate is a constraint on procedures using the diagonal. A finite scan over product directions of coordinate subtensors can give shorter expected length at zero while preserving the original uniform length bound. The scan is an existence construction with exponential complexity; no efficient implementation is asserted.

For a tensor B, define its injective norm

$$\rho(B) = \sup_{\|u_k\|_2=1} |\langle B, u_1 \otimes \cdots \otimes u_K\rangle|.$$

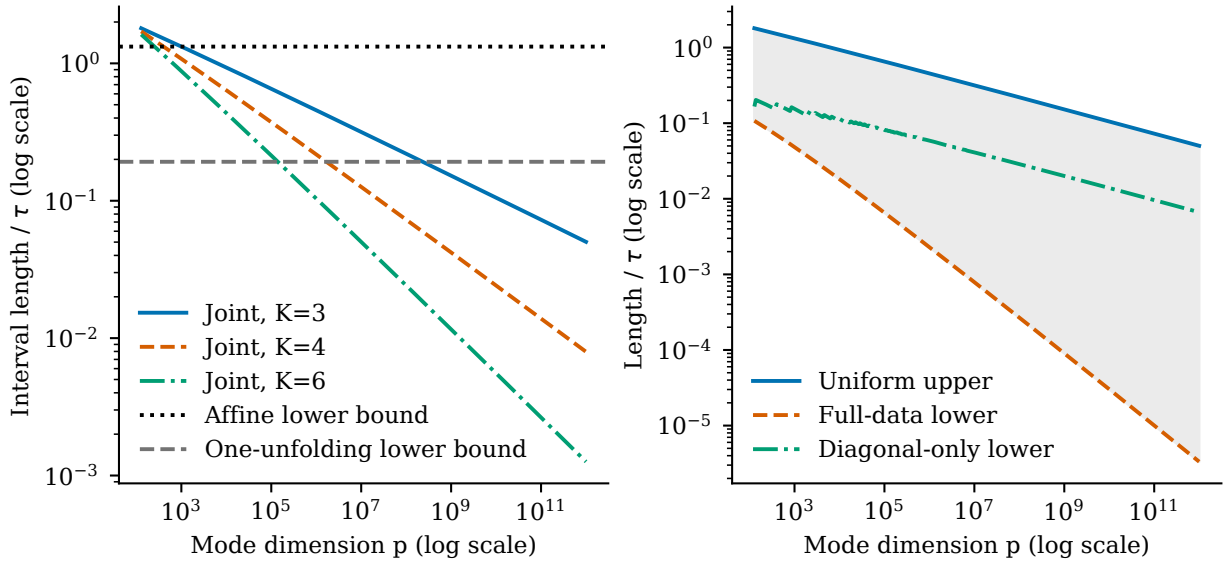


Figure 1: Joint constraints permit vanishing normalized length, and the diagonal-only lower bound matches its order. Left: cap-one selected length and the affine and one-unfolding lower bounds, for $M = \tau\sqrt{p}$ and $\alpha = 0.05$. Right: the $K = 3$ upper bound, the full-data lower bound and the diagonal-only lower bound from (23), using $s = \lfloor (p/\log p)^{1/3} \rfloor$. The right panel shows the initial bounds, before the improvements in Theorems 7–10; it is not the final full-data bracket. Curves evaluate analytic formulas up to $p = 10^{12}$; simulated dimensions stop at 8192.

For every nonempty subset $S \subseteq \{1, \dots, p\}$ of size s, choose fixed $1/(2K)$-nets of the unit sphere in $\mathbb{R}^s$, one per mode, each with at most $(1 + 4K)^s$ points. Let $N_S(Z)$ be the largest absolute contraction of $Z_{S,\dots,S}$ over their product. Define

$$\lambda_s^{\mathrm{scan}} = \tau \left[2\{s \log(ep/s) + Ks \log(1 + 4K) + \log(2p/\gamma)\}\right]^{1/2}, \tag{25}$$

$$R_\gamma = \frac{4J^{K/2}}{\sqrt{p}} \sum_{s=1}^{p} \lambda_s^{\mathrm{scan}} s^{-K/2}. \tag{26}$$

Accept the zero test if $N_S(Z) \le \lambda_{|S|}^{\mathrm{scan}}$ for every nonempty S. On acceptance, let $C_{\mathrm{scan}} = [-h_{\mathrm{scan}}, h_{\mathrm{scan}}]$, where $h_{\mathrm{scan}} = \min(h_0, R_\gamma)$; on rejection let $C_{\mathrm{scan}} = [-h_0, h_0]$. Here $h_0 = M\sqrt{s_{\mathrm{cap}}/p}$ and s_{cap} is the smallest cap, truncated at p. For $0 < \alpha_0, \gamma < 1$ with $\alpha_0 + \gamma \le \alpha$, intersect this interval with $C_*(\alpha_0)$ and call the result $C_\dagger$; replace an empty intersection by $\{0\}$. Write $\ell_* = |C_*(\alpha_0)|$.

Theorem 7 (A full-data interval with short expected length at zero)**.** *Under* (1), (16) *and either cap class,* C_{scan} *is honest at level* $1 - \gamma$*, and* $C_\dagger$ *is honest at level* $1 - \alpha$*. Moreover,*

$$\sup_{A \in \Theta} \mathbb{E}_A |C_\dagger| \le \ell_*, \qquad \mathbb{E}_0 |C_\dagger| \le \min\{\ell_*, 2R_\gamma + \ell_* \gamma\}. \tag{27}$$

For fixed $0 < \alpha < 1/4$*, fixed* $K \ge 3$*, fixed positive caps and* $M = \tau\sqrt{p}$*, choose* $\alpha_0 = \alpha/2$ *and* $\gamma = \alpha/(2p)$*. Then*

$$\mathbb{E}_0 |C_\dagger| \le C_{K,J,\alpha} \frac{\tau\sqrt{\log p}}{\sqrt{p}} \sum_{s=1}^{p} s^{-(K-1)/2} \tag{28}$$

for sufficiently large p. Consequently,

$$\mathcal{L}_0(\Theta) \asymp \tau\sqrt{\frac{\log p}{p}}, \qquad K \geq 4. \tag{29}$$

The matching lower bound in (29) is Theorem 6, applied to the same full-data experiment and honesty class. The sum in (28) is bounded for $K \geq 4$ and harmonic for $K = 3$. The number of scanned product directions is bounded by $(1 + (1 + 4K)^K)^p - 1$. Thus the result establishes attainable statistical precision, without a polynomial-time guarantee. The worst-case bound in (27) is the initial diagonal upper order. The residual projection in Theorem 10 strengthens it to the sharp full-data rate.

Theorem 8 (A harmonic obstruction in order three)**.** *Under (1) and (16), let $K = 3$, $M = \tau\sqrt{p}$ and $0 < \alpha < 1/4$. Every interval honest on a class containing the rank-one tensor Frobenius ball satisfies*

$$\mathbb{E}_0|C| \geq (3/4 - 2\alpha)\frac{\tau H_p}{200\sqrt{p}}, \qquad H_p = \sum_{i=1}^{p} i^{-1}. \tag{30}$$

Theorem 8 uses a full-data mixture of rank-one tensors with sign-coupled factors and a common harmonic diagonal. It refutes the earlier conjecture that $\mathcal{L}_0$ always has order $\tau\sqrt{\log p/p}$ at rank one. Combining (30) and (28) gives the intermediate cubic bracket

$$c_\alpha^{\mathrm{cub}} \frac{\tau \log p}{\sqrt{p}} \leq \mathcal{L}_0(\Theta) \leq C_{J,\alpha}\frac{\tau(\log p)^{3/2}}{\sqrt{p}}. \tag{31}$$

Here $c_\alpha^{\mathrm{cub}} = (3/4 - 2\alpha)/200 > 0$. The intermediate bracket (31) has a factor $\sqrt{\log p}$ gap. Theorem 9 closes it from below. The constant $1/200$ is conservative and is not an optimal finite-sample constant.

Further verified refinements. The accompanying theory review proves a smaller rank coefficient in the diagonal bound, an exact one-dimensional characterization of the optimal fixed affine radius, and a noise-envelope version with an additive approximation term. It also records finite-variance and unknown-Gaussian-variance extensions requiring retained replicates. These supporting results are collected in Appendix C; the simulations retain the original Gaussian threshold procedure.

5.6 Sharp full-data rates, including worst-case length

A sparse harmonic profile strengthens the cubic lower bound. Inverting the residual scan then turns the zero-point construction into an interval with uniformly controlled length.

Theorem 9 (Sparse harmonic lower bound)**.** *Under (1) and (16), let $K = 3$ and $0 < \alpha < 1/4$. For every integer $1 \leq m \leq p$, put*

$$b_m = \min\left\{\frac{M}{\sqrt{27m}}, \frac{\tau}{\sqrt{27}}\sqrt{\log(1 + p/(4m^2))}\right\}, \qquad c_{\alpha,d} = 1 - 2\alpha - \tfrac{1}{2}\sqrt{e^{1/4} - 1}.$$

Every interval honest on a class containing the rank-one Frobenius ball satisfies

$$\mathbb{E}_0|C| \geq c_{\alpha,d}\frac{b_m H_m}{\sqrt{p}}. \tag{32}$$

If $M = \tau\sqrt{p}$, then for $p \geq 256$ and $m = \lfloor p^{1/4} \rfloor$,

$$\mathbb{E}_0|C| \geq \frac{c_{\alpha,d}}{16\sqrt{27}}\frac{\tau(\log p)^{3/2}}{\sqrt{p}}.$$

The profile weights, rather than additional rank components, produce the harmonic sum. The alternatives are nonnegative rank-one tensors. Their full likelihood overlap is bounded linearly in the support intersection, permitting the classical hypergeometric calculation.

Write $\lambda_s(\gamma) = \lambda_s^{\mathrm{scan}}$ for the threshold in (25) at error budget γ. Use the product nets above with thresholds $\lambda_s(\alpha)$. Define

$$\mathcal{F}(Z) = \{B \in \Theta(M, \bar{r}) : N_S(Z - B) \le \lambda_{|S|}(\alpha) \text{ for every nonempty } S\}.$$

Let C_F be the interval between the minimum and maximum of ψ on $\mathcal{F}(Z)$, and set $C_F = \{0\}$ when this set is empty. Write $J_2 = J(2\bar{r})$, the expansion constant with doubled caps, and put

$$W_\alpha = \frac{4J_2^{K/2}}{\sqrt{p}} \sum_{s=1}^{p} \lambda_s(\alpha) s^{-K/2}.$$

Theorem 10 (Uniform diameter and full-data minimax length)**.** *Under (1), (16), $0 < \alpha < 1$ and either positive cap class, C_F is measurable and honest at level $1 - \alpha$, and for every observation*

$$|C_F| \le \min(2h_0, W_\alpha). \tag{33}$$

For fixed $0 < \alpha < 1/4$, fixed $K \ge 3$, fixed positive caps and $M = \tau\sqrt{p}$,

$$\mathcal{L}_0(\Theta) \asymp \mathcal{L}_{\max}(\Theta) \asymp r_K(p), \qquad r_K(p) = \begin{cases} \tau(\log p)^{3/2}/\sqrt{p}, & K = 3, \\ \tau\sqrt{\log p/p}, & K \ge 4. \end{cases} \tag{34}$$

Constants may depend on K, the caps and α.

Differences of feasible tensors obey doubled unfolding caps; the proof does not assume that a rank class is closed under subtraction. Computing the confidence-set extrema is not claimed to be efficient.

Proposition 2 (Squared-error consequence)**.** *In the regime of (34),*

$$\inf_{\widehat{\psi}} \sup_{A \in \Theta} \mathbb{E}_A(\widehat{\psi} - \psi(A))^2 \asymp r_K(p)^2.$$

The midpoint of C_F with error level $\alpha_p = p^{-2}$ attains the upper order for $p \ge 2$.

5.7 Computation: a certified hierarchy and a restricted lower bound

The exact residual projection supplies the statistical benchmark. A truncated zero test is computable with an explicit search budget. It does not compute the extrema defining C_F.

Proposition 3 (Truncated scans)**.** *For $1 \le t \le p$, test zero using only supports of sizes at most t, retaining thresholds calibrated simultaneously over all sizes. Write $\lambda_s = \lambda_s(\gamma)$ and put*

$$q = 2/K, \quad B_q = J^{1-1/K} M^q, \quad a_t = 4J^{K/2}\lambda_t t^{-K/2},$$

$$R_t = \frac{4J^{K/2}}{\sqrt{p}} \sum_{s=1}^{t} \lambda_s s^{-K/2} + \mathbf{1}_{\{t<p\}} \frac{B_q}{\sqrt{p}} a_t^{1-q}.$$

On acceptance use radius $\min(h_0, R_t)$ *centered at zero; on rejection use the structural interval. Intersect with* $C_*(\alpha_0)$*, where* $0 < \alpha_0, \gamma < 1$*,* $\alpha_0 + \gamma \le \alpha$*, and* γ *is the scan error budget, using the singleton convention. The resulting* C_t *is honest, has length at most* $\ell_* = |C_*(\alpha_0)|$*, and*

$$\mathbb{E}_0|C_t| \le \min(\ell_*, 2R_t + \ell_*\gamma).$$

For fixed $K \ge 3$*, fixed caps and fixed* $0 < \alpha < 1$*, with* $M = \tau\sqrt{p}$*,* $\alpha_0 = \alpha/2$ *and* $\gamma = \alpha/(2p)$*,*

$$\begin{aligned} R_t \lesssim \frac{\tau\sqrt{\log p}}{\sqrt{p}} \sum_{s=1}^{t} s^{-(K-1)/2} \\ + \mathbf{1}_{\{t<p\}}\tau(\log p/p)^{\beta} t^{-(K-1)\beta}, \qquad \beta = (K-2)/(2K). \end{aligned} \tag{35}$$

Explicit normalized grid nets have at most $(3 + 4K\sqrt{s})^s$ directions per mode. Using their cardinalities in the Gaussian thresholds preserves (35). The arithmetic cost through level t is bounded by

$$O\left\{\sum_{s=1}^{t} \binom{p}{s} (3 + 4K\sqrt{s})^{Ks} s^K\right\}.$$

For fixed t, K this is polynomial in p. Fixed t does not close the power gap in the tail term of (35). Completed levels share one noise event, so successive intersections remain honest even if the decision to compute another complete level uses the data.

A complementary obstruction holds for polynomial estimators under squared-error loss. This is the restricted low-degree framework used in sparse tensor PCA [Choo and d'Orsi, 2021, Fu and Sohn, 2026]; it does not characterize all polynomial-time computation.

Theorem 11 (A low-degree estimation separation)**.** *Let* $K \ge 3$*,* $1 \le s \le \sqrt{p}$*, and mix uniformly over*

$$x_S = s^{-1/2}\mathbf{1}_S, \qquad A_S = \tau\lambda x_S^{\otimes K}, \qquad |S| = s.$$

Assume $0 < \tau\lambda \le M$*, and write*

$$\delta = \frac{\tau\lambda s^{1-K/2}}{\sqrt{p}}, \quad \mu = s^2/p, \quad B_D = \frac{\lambda^2}{s^K}(KD)^K,$$

where $D \ge 1$ *is the polynomial degree. If* $B_D < 1$*, the centered likelihood ratio projected onto polynomials of degree at most* D *has squared null norm at most*

$$\eta_D^2 = \mu e^{\mu} B_D/(1 - B_D).$$

Every scalar polynomial estimator $T(Z)$ *of degree at most* D *satisfies*

$$\sup_{A\in\Theta} \mathbb{E}_A(T - \psi(A))^2 \ge \frac{\delta^2}{(2 + \eta_D)^2} \tag{36}$$

on any parameter class containing zero and these alternatives. In particular, let $M = \tau\sqrt{p}$*,* $\lambda = \sqrt{p}$*,* $s = \lfloor p^a \rfloor$*, and* $D = \lfloor p^b \rfloor$*, where*

$$\frac{1}{K} < a < \min\left(\frac{1}{2}, \frac{1}{K-2}\right), \qquad 0 < b < a - \frac{1}{K}.$$

Then $\eta_D \to 0$ *and the lower order in* (36) *is polynomially larger than the unrestricted rate in Proposition 2.*

This theorem includes $K = 3$ and uses nonnegative sparse factors. It does not imply the same limitation for arbitrary interval endpoints or arbitrary thresholded algorithms.

Proposition 4 (What an efficient short interval would imply)**.** *Let Q be supported on parameters with common target $\delta > 0$, and let the null target be zero. An interval honest at level $1 - \alpha$ at the null and these alternatives defines $\phi(Z) = \mathbf{1}\{\delta \in C(Z)\}$ with*

$$Q(\phi = 0) \le \alpha, \qquad P_0(\phi = 1) \le \alpha + \mathbb{E}_0|C|/\delta.$$

A computational hypothesis excluding polynomial-time tests with total error below $c > 2\alpha$ therefore excludes polynomial-time honest intervals with $\mathbb{E}_0|C| < (c - 2\alpha)\delta$ for that prior.

The last implication is conditional on a testing-hardness hypothesis. Theorem 11 alone does not establish that hypothesis. Efficient minimax confidence intervals remain open, with a proved restricted estimation obstruction and an explicit computable hierarchy.

6 Norm budgets and loading geometry

The restriction $M = \tau\sqrt{p}$ can be removed for the equal diagonal contrast. The key is to impose the Frobenius budget and the residual-scan constraints on each ordered diagonal entry, before summing. Throughout this section the tensor order and positive rank caps are fixed; constants are not claimed uniform in growing caps.

Put $P_2 = 2J_2^{(K-1)/2}$ and $Q_2 = 4J_2^{K/2}$, where $J_2 = J(2\bar{r})$, and define

$$U_{K,p}(M;\alpha) = \frac{1}{\sqrt{p}} \sum_{s=1}^{p} \min\{P_2 M, Q_2 \lambda_s(\alpha)\}\, s^{-K/2}. \tag{37}$$

Here $\lambda_s(\alpha)$ is the finite-net threshold in (25), with error budget α.

Theorem 12 (Diameter using both norm and scan constraints)**.** *For every $p \ge 1$, $K \ge 3$, $M, \tau > 0$ and $0 < \alpha < 1$, the residual-projection interval C_F is honest and satisfies, for every observation,*

$$|C_F| \le \min\{2h_0, U_{K,p}(M;\alpha)\}.$$

The bound retains the supplied norm budget inside the ordered-coordinate sum; taking the minimum only after summation can lose the intermediate cubic regime.

Let $\ell_p = \log(ep)$, $x = M/(\tau\sqrt{\ell_p})$, and define

$$g_p(x) = \min\{x, 1 + \log_+ x, \ell_p\}, \qquad \log_+ x = \max(0, \log x), \tag{38}$$

$$\mathfrak{r}_K(p, M, \tau) = \frac{\tau\sqrt{\ell_p}}{\sqrt{p}} \begin{cases} g_p(x), & K = 3, \\ \min(x, 1), & K \ge 4. \end{cases} \tag{39}$$

These definitions also cover $p = 1$, when the rate is $\min(M, \tau)$.

Theorem 13 (Full-data length for every positive norm budget)**.** *Fix $K \ge 3$, positive Tucker or TT caps, and $0 < \alpha < 1/4$. For the equal diagonal contrast there are constants $0 < c < C < \infty$, depending only on these fixed quantities, such that for every $p \ge 1$ and every $M, \tau > 0$,*

$$c\,\mathfrak{r}_K(p, M, \tau) \le \mathcal{L}_0(\Theta) \le \mathcal{L}_{\max}(\Theta) \le C\,\mathfrak{r}_K(p, M, \tau).$$

The lower bounds already hold when the honesty class is restricted to rank-one tensors. The interval in Theorem 12 attains the upper order.

For $K \geq 4$ the rate is $\min\{M, \tau\sqrt{\log(ep)}\}/\sqrt{p}$. For $K = 3$, the norm-limited regime $x \leq 1$ has rate $M/\sqrt{p}$; above it the multiplier grows as $1 + \log x$ until capped at ℓ_p. These are rates up to fixed multiplicative constants, not exact transition constants. Substituting $M = \tau\sqrt{p}$ recovers Theorem 10.

For squared-error estimation, a fixed-confidence midpoint argument is insufficient when M/τ is arbitrarily large: its off-coverage term can grow with M^2. Instead use an approximate minimum-residual estimator. Fix $\beta = 1/8$, and put

$$T(R) = \max(0, \max_{S \neq \varnothing}\{N_S(R) - \lambda_{|S|}(\beta)\}).$$

Choose a fixed countable dense sequence (B_j) in the compact cap class. Define

$$d(Z) = \inf_j T(Z - B_j), \quad j(Z) = \min\{j : T(Z - B_j) \leq d(Z) + \tau\}, \quad \widehat{\psi} = \psi(B_{j(Z)}). \tag{40}$$

The tolerance τ ensures the displayed index exists. This defines a statistical estimator, not a claimed efficient optimization algorithm.

Theorem 14 (Squared risk uniformly over norm budgets)**.** *Under the fixed-order, fixed-cap conditions of Theorem 13,*

$$\inf_{\widetilde{\psi}} \sup_{A \in \Theta} \mathbb{E}_A(\widetilde{\psi} - \psi(A))^2 \asymp \mathfrak{r}_K(p, M, \tau)^2$$

uniformly over $p \geq 1$ and $M, \tau > 0$. The measurable estimator in (40) attains the upper bound. It uses neither a failure probability depending on M nor a minimum nonzero singular value.

6.1 Nonuniform orthogonal loadings

Return to the prescribed orthogonal product loading in (4), with m nonzero weights and possibly unequal ambient mode dimensions. Let w_s^* denote decreasing absolute weights. Use orthogonal mode transformations and restrict to the m^K coordinates spanned by the prescribed loading vectors. The resulting Gaussian tensor retains the rank and norm bounds. Write $\lambda_s^{(m)}$ for the scan threshold with $p = m$, and define

$$U_w = \sum_{s=1}^{m} w_s^* \min\{P_2 M, Q_2 \lambda_s^{(m)}(\alpha)\} s^{-K/2}, \tag{41}$$

$$L_w = \max_{1 \leq d \leq m} w_d^* \sqrt{d}\, \mathfrak{r}_K(d, M, \tau). \tag{42}$$

The rank constants use the same supplied caps. The structural radius is

$$h_0 = M \left(\sum_{s=1}^{s_{\mathrm{cap}}} (w_s^*)^2 \right)^{1/2},$$

with s_{cap} truncated at m.

Theorem 15 (Loading-dependent full-data bounds)**.** *For fixed $K \geq 3$, fixed positive caps and fixed $0 < \alpha < 1/4$, an honest interval obtained by residual projection in the rotated block satisfies*

$$|C_w(Z)| \leq \min(2h_0, U_w)$$

for every observation. Conversely every interval honest on the full ambient cap class satisfies

$$\mathbb{E}_0|C(Z)| \geq c_{K,\alpha} L_w.$$

The lower bound allows use of every ambient tensor entry. These bounds are not asserted to match for arbitrary nonuniform weights.

Proposition 5 (Sharp rates for a flat orthogonal loading)**.** *If $w_i = \epsilon_i/\sqrt{m}$ with prescribed $\epsilon_i \in \{-1, 1\}$, then both minimax expected-length criteria have order $\mathfrak{r}_K(m, M, \tau)$, and minimax squared error has its square, independently of the larger ambient mode dimensions. The statement includes dense orthogonal rotations of the loading. At $m = 1$ it gives the bounded-normal-mean scale* $\min(M, \tau)$.

What remains unresolved. The norm/noise restriction is removed for equal or flat orthogonal loadings. General weights now have explicit upper and lower envelopes; an arbitrary tensor loading without the orthogonal product representation remains untreated. Efficient minimax computation, sharp growing-cap dependence, and adaptation to unknown actual rank beyond the supplied-cap guarantees remain open. In particular, honesty on any class containing the rank-one ball forces positive expected length at zero of order (39); the zero-length oracle available when rank zero is known cannot be attained under that outer honesty requirement.

7 Robust noise, computation, and rank adaptation

This section separates extensions of validity from sharp minimax claims. It also separates adaptation on a spectrally separated class from honesty on the original class containing arbitrarily weak components. Lower-bound comparisons use a fixed error level below 1/4. All statements have proofs in Appendix F.

7.1 Beyond Gaussian observations

For a centered error tensor E, the following joint directional condition is sufficient:

$$\mathbb{E}\exp(t\langle U, E\rangle) \le \exp(t^2 v^2 \|U\|_F^2/2) \quad (U \text{ deterministic}, \ t \in \mathbb{R}). \tag{43}$$

It allows dependence and nonidentical coordinates. Independent centered sub-Gaussian coordinates with common proxy v satisfy it, as does Gaussian covariance bounded above by $v^2 I$. Marginal coordinate bounds alone do not imply it.

Theorem 16 (Full-data robustness under directional tails)**.** *In $Z = A + E$, replace τ by a known $v > 0$ in the scan thresholds and residual-fit tolerance. Under (43), the norm-sensitive diameter, uniform squared-risk upper bound, and orthogonal-loading upper bounds remain valid. For fixed order/caps, their equal-diagonal orders are $\mathfrak{r}_K(p, M, v)$ and its square. If the noise law is an unknown member of the family satisfying (43), these rates are minimax when coverage and the supremum risk/length range over that family. This lower-bound statement need not hold for each fixed non-Gaussian law separately.*

For a finite-variance extension retain independent identically distributed full-tensor replicates $W_1, \dots, W_n$, with mean A and $\mathrm{Cov}(\mathrm{vec}\, W_j) \preceq \sigma^2 I$ for known $\sigma > 0$. Put $\tau = \sigma/\sqrt{n}$, and let

$$H_s = s\log(ep/s) + Ks\log(1 + 4K).$$

For a desired error $0 < \beta < 1/4$, choose b_s as the smallest odd integer at least

$$\max\{9, [H_s + \log(p/\beta)]/\log 2\}, \qquad n \ge 2\max_s b_s. \tag{44}$$

For each unit product-net direction U supported on S^K, $|S| = s$, let d_U be the median of its b_s disjoint block averages, using block size $k_s = \lfloor n/b_s \rfloor$. The block partitions may differ across s but are deterministic. Set $\lambda_s^{\mathrm{mom}} = 4\sigma/\sqrt{k_s}$.

Theorem 17 (Full scans with only finite-variance replicates)**.** *Use the feasible constraints* $|d_U - \langle U, B\rangle| \le \lambda_s^{\mathrm{mom}}$ *in place of the Gaussian residual constraints. Their projection onto the equal-diagonal target is honest at level* $1-\beta$*, with the norm-sensitive diameter using* λ_s^{mom}*. For fixed* K*, caps and confidence, both minimax expected-length criteria, with a supremum over all the stated replicate laws, have order* $\mathfrak{r}_K(p, M, \sigma/\sqrt{n})$ *whenever* (44) *holds.*

For squared error use $\beta = 1/8$ *and approximately minimize*

$$F(B) = \max(0, \max_U\{|d_U - \langle U, B\rangle| - \lambda_{s(U)}^{\mathrm{mom}}\})$$

over a fixed countable dense cap-class sequence, within tolerance τ*. The resulting scalar estimator has minimax squared risk of order* $\mathfrak{r}_K^2$ *over the same family. This is an existence construction. The sufficient replicate count is* $n \ge C_{K,\beta}p$*, and the full directional search remains exponential.*

The moments d_U need not be contractions of a single tensor estimate. Their separate linear constraints suffice. No Gaussian likelihood is used for the robust upper bounds; the lower bounds use the Gaussian subfamily of replicate distributions. This is a finite-direction use of the classical median-of-means principle [Devroye et al., 2016].

Proposition 6 (Two limits of a noise relaxation)**.** *(i) In one dimension, known noise taking values* $\pm v$ *with equal probabilities permits exact recovery of every mean* $|A| \le M < v$*, although it has variance* v^2 *and satisfies* (43)*. Thus a Gaussian lower bound cannot be transferred to every noise law from a variance bound alone. (ii) For* $E = v\xi\mathbf{1}$ *with one Rademacher sign* ξ*, every coordinate has proxy* v*, but a flat unit product contraction on a* p^K *block has variance* v^2p^K*. Coordinatewise calibration cannot certify a full scan or a dense rotated loading.*

7.2 Approximate ranks and growing caps

Theorem 18 (Sharp length under a Frobenius approximation allowance)**.** *In the original Gaussian experiment let* $\Theta^\eta = \Theta(M, \bar{r}) + \eta\mathbb{B}_F$*,* $\eta \ge 0$*. For the equal diagonal, fixed* $K \ge 3$*, fixed positive caps and fixed* $0 < \alpha < 1/4$*,*

$$\mathcal{L}_0(\Theta^\eta) \asymp \mathcal{L}_{\max}(\Theta^\eta) \asymp \min\{\tau, \mathfrak{r}_K(p, M, \tau) + \eta\}.$$

The upper interval inflates every scan threshold by η *and both projected endpoints by* η*, with a deterministically selected direct Gaussian fallback. Thus approximate low rank is allowed, but the original rate requires* $\eta = O(\mathfrak{r}_K)$*.*

Proposition 7 (A growing-rank lower bound and the unrestricted endpoint)**.** *Let* r_* *be the minimum supplied cap, truncated at* p*. Under equal loading every honest interval obeys*

$$\mathbb{E}_0|C| \ge \frac{c_\alpha}{\sqrt{p}} \max_{1\le s\le r_*} \min\{M\sqrt{s}, \tau s\sqrt{\log(1 + p/(4s^2))}\}.$$

The finite upper envelopes of the preceding sections remain valid for dimension-dependent supplied caps with their displayed cap constants; these constants are not uniformly bounded. If all rank restrictions are removed, then for any prescribed unit loading both length criteria have order $\min(M, \tau)$*, attained in polynomial time. The vanishing tensor rate therefore does not persist under unrestricted ranks.*

7.3 Arbitrary loading tensors and efficient regimes

For a nonzero, otherwise arbitrary deterministic tensor L, a useful sufficient description is a certified decomposition $L = L_0 + R$, where L_0 has a tractable orthogonal loading representation and the remainder R is known exactly. Normalization is not required in this subsection.

Proposition 8 (A confidence interval for any certified loading decomposition)**.** *Suppose $C_0 = [a_0, b_0]$ is honest for $\langle L_0, A\rangle$ at error α_0, and $\mathbb{P}(|\langle R, E\rangle| > q_R) \le \alpha_1$ uniformly. Then*

$$C_L = [a_0 + \langle R, Z\rangle - q_R,\ b_0 + \langle R, Z\rangle + q_R]$$

is honest at error $\alpha_0 + \alpha_1$ and has length $|C_0| + 2q_R$. Under (43), one may take

$$q_R = v\|R\|_F\sqrt{2\log(2/\alpha_1)}.$$

For Gaussian noise use the normal quantile. No independence between the two terms is required. A decomposition or partition selected from loading and supplied constants alone needs no selection penalty. Selection using Z does require simultaneous calibration.

This also supplies additional unequal-weight upper bounds: partition the orthogonal terms into prescribed blocks, apply their separate weighted bounds with error budgets summing to α, and add the intervals. Include the unsplit construction among the choices. The minimum among their deterministic diameter bounds is achievable by choosing the construction before observing data.

Theorem 19 (Two efficiently attained minimax regimes)**.** *(i) For the equal diagonal, fixed order/caps and $M \le \tau\sqrt{\log(ep)}$, the structural interval is minimax up to fixed constants, with length of order $M/\sqrt{p}$. (ii) Let a parameter class contained in $\{A : \|A\|_F \le M\}$ contain every rank-one tensor of Frobenius norm at most M. If a supplied unit product tensor U certifies $|\langle L, U\rangle| \ge c\|L\|_F$ for a fixed $c > 0$, then under Gaussian observations both length criteria have order $\|L\|_F \min(M, \tau)$. Choose between the norm-only structural interval and the direct Gaussian interval. The algorithm uses a contraction and scalar comparisons, without rank recovery or tensor optimization. This covers arbitrary loadings with a certified constant product overlap, regardless of supplied rank caps.*

The general loading transfer is a valid upper bound, not a characterization of the optimum for every tensor L. The near-flat difficult contrasts do not satisfy a dimension-independent product-overlap condition.

Proposition 9 (Why full convexification loses the weak-signal gain)**.** *Any convex parameter set containing the rank-one Frobenius ball of radius M also contains $\{tL : |t| \le M/\sqrt{p}\}$ for the equal diagonal loading. Consequently, every interval honest on that convex set satisfies*

$$\mathbb{E}_0|C| \ge c_\alpha \min(M/\sqrt{p}, \tau).$$

At $M = \tau\sqrt{p}$ this is order τ, exceeding the original full-data rate by an unbounded factor as $p \to \infty$. The conclusion concerns procedures guaranteeing coverage on the whole convex outer class; it is not a computational lower bound for every algorithm using a convex subroutine.

7.4 Actual rank: an obstruction and a computable separated case

Theorem 20 (An adaptation obstruction at nonzero tensors)**.** *Suppose A_0 is supported on $\{1, \dots, q\}^K$, $q < p$, has norm $a < M$, and every supplied unfolding cap is at least its actual*

rank plus one. Put $m = p - q$ and $M_0 = (M^2 - a^2)^{1/2}$. For the equal diagonal every interval honest on the outer cap class satisfies

$$\mathbb{E}_{A_0}|C| \geq c_{K,\alpha}\sqrt{m/p}\,\mathfrak{r}_K(m, M_0, \tau).$$

Thus arbitrarily weak extra components obstruct shortening at some nonzero lower-rank tensors, not only at zero. This is a lower bound, not a complete characterization of adaptation on all rank subclasses.

An explicit positive result uses two independent observations $Z^{(j)} = A + E^{(j)}$, with directional proxies v_j as in (43). Let $d_k = \prod_{\ell \neq k} p_\ell$, choose $0 < \delta < \alpha$, and set

$$\eta_k = 2v_1\sqrt{2\{(p_k + d_k)\log 9 + \log(2K/\delta)\}}. \tag{45}$$

The allowed class has $\|A\|_F \leq M$ and, in each mode, either rank zero or smallest nonzero singular value strictly greater than $2\eta_k$. No ranks or rank caps are supplied to the algorithm. Separation is a model assumption; observed singular values alone cannot certify the absence of undetected weak components. Compute $\widehat{r}_k = \#\{j : \sigma_j(Z^{(1)}_{(k)}) > \eta_k\}$ and let Q_k project onto those left singular vectors. If any $\widehat{r}_k = 0$, return $\{0\}$. Otherwise put

$$d_k^* = \min\left\{1, \frac{2\eta_k}{\sigma_{\widehat{r}_k}(Z^{(1)}_{(k)}) - \eta_k}\right\}.$$

Here d_k^* is an error bound, distinct from the unfolding column dimension d_k. For the commuting mode projectors define the orthogonal projector

$$T = \bigotimes_k Q_k + \sum_k (I - Q_k) \otimes \bigotimes_{\ell \neq k} Q_\ell.$$

The tensor factors in this expression occupy their respective modes. This first-order product-space projector may be larger than the exact Tucker manifold tangent projector at ranks above one. Set

$$B_L = M\|(I - T)L\|_F \min\left\{1, \left(\sum_{k<\ell}(d_k^* d_\ell^*)^2\right)^{1/2}\right\}, \tag{46}$$

$$h_T = B_L + v_2\|TL\|_F\sqrt{2\log(2/(\alpha - \delta))}. \tag{47}$$

Theorem 21 (Finite-sample rank calibration with a computable bias correction)**.** *The interval centered at $\langle TL, Z^{(2)}\rangle$ with radius h_T is honest at level $1 - \alpha$ on the separated class just specified, for any prescribed loading L. One may choose, using only the pilot and known quantities, the shortest-radius option among it, the direct second-observation interval with error $\alpha - \delta$, and the norm-only structural interval. It may also be intersected with the structural interval. For Gaussian second-observation noise the square-root logarithm can be replaced by $z_{1-(\alpha-\delta)/2}$.*

With probability at least $1 - \delta$, all selected mode ranks equal the actual ranks and (46) bounds the functional bias. The method consists of unfolding SVDs and tensor projections and is polynomial in the explicit input size. Its noise radius depends on $\|TL\|_F$, and its additional cost is the displayed bias term. General minimax optimality and adaptation without the spectral separation are not asserted.

For one Gaussian observation, independent artificial G' gives $Z^{(1)} = Z + \tau G'$ and $Z^{(2)} = Z - \tau G'$, with independent errors of scale $\sqrt{2}\tau$. For non-Gaussian errors, this trick need not give independence; genuinely independent replicate groups suffice. The rank statement concerns Tucker mode ranks, and applies to TT tensors meeting the mode-gap conditions, but does not identify their TT bond ranks. Subspace projection is established methodology [Xu et al., 2025]; the present contribution is an explicit conservative finite-sample certificate in this setting.

Assumption audit. Gaussian isotropy gives the exact diagonal sequence and the likelihood-ratio identities. The threshold interval alone extends whenever a valid simultaneous coordinate noise envelope is supplied, but the direct alternative and lower bounds require their own calibration. Orthonormal loading families produce Bessel bounds and equal noise variances; arbitrary loadings are untreated. Canonical tensor expansions avoid a spectral gap but yield potentially large rank constants. Supplied caps and M identify the outer honesty class; replacing them with unvalidated estimates is not covered. None of these assumptions should be described as a routine approximation for dependent LLM representations.

8 Simulation study

8.1 Protocol and baselines

The simulation protocol was written after the theorem derivations and before implementation. It uses seed 20260909, 2,000 replications per configuration, and exact marginal observations of the diagonal coordinates needed by the procedures. Unused independent Gaussian coordinates of the full tensor need not be generated. This does not simulate a regression design or a tensor completion experiment.

The main grid takes $K \in \{2, 3, 4, 6\}$, $p = n \in \{128, 512, 2048, 8192\}$, $M = \sigma = 1$, and rank cap one. Signals have the form $A = au^{\otimes K}$, with $a \in \{0, 0.02, 0.2, 1\}$ and u uniform on a coordinate support. For nonzero signals, support sizes include one, p, and a value near $(a/\lambda)^{2/K}$, with duplicates removed. The latter places diagonal means near the threshold. Additional configurations vary n/p and the valid threshold multiplier, and use cap two with a second component tending to zero. There are 175 configurations, of which three deliberately violate the iid Gaussian model.

We compare I_G, I_0, I_{NL} and C_*. For nonzero rank-one signals, a known-direction oracle is also included. If $U = u^{\otimes K}$ were supplied, then $c\langle U, Z\rangle \pm z\tau|c|$, where $c = \langle L, U\rangle$, is exact for $A = aU$. Its joint Gaussian noise with the diagonal is simulated exactly. This oracle has unavailable structural information and a smaller honesty class; it is not an implementation of an estimated-subspace method. Published strong-signal methods are compared through their assumptions rather than advertised as uniform baselines without a matching implementation.

Each output row reports coverage, binomial Monte Carlo standard error, an exact 95% Clopper–Pearson interval, deterministic length, center bias and its standard error, and root mean squared center error. Width standard errors are zero because the radii are deterministic. Numerical proof checks separately examine 101,101 scalar soft-threshold cases, 120 random Tucker and 120 random TT tensors, and exact finite-binomial mixture second moments. The diagonal benchmark is additionally checked against an enumeration of 3,136 support pairs and exact hypergeometric moments in 24 configurations. They supplement rather than replace the proofs.

8.2 Gaussian coverage and interval length

Table 3 summarizes the 172 in-model configurations. The selected interval is substantially conservative. Across configurations, direct Gaussian coverage ranges from 0.9365 to 0.9625, and selected coverage ranges from 0.9975 to 1. These extrema are descriptive, not simultaneous confidence limits for uniform coverage. At coverage 0.95, the Monte Carlo standard error with 2,000 replications is approximately 0.00487. When all replications cover, the exact binomial interval is approximately $[0.99816, 1]$, despite a zero plug-in standard error.

Method	Configurations	Min. coverage	Max. coverage
Direct Gaussian	172	0.9365	0.9625
Structural	172	1.0000	1.0000
Threshold	172	0.9975	1.0000
Selected	172	0.9975	1.0000
Known-direction oracle	117	0.9370	0.9615

Table 3: Coverage extrema over the stated Gaussian configurations. The oracle is evaluated only for nonzero rank-one signals, giving fewer configurations and a different information set. Each configuration has 2,000 replications; configuration-level uncertainty is retained in the accompanying CSV.

At $p = n = 8192$ and cap one, the selected lengths divided by τ are 2.0000, 0.9622, 0.6674 and 0.4629 for $K = 2, 3, 4, 6$, respectively. The direct length is 3.9199τ. These width values are exact evaluations of the formulas, not fitted empirical convergence rates. They agree with the order-dependent prediction, while the $K = 2$ control shows no vanishing length ratio. A known-direction oracle can be much shorter for diffuse signals; there is no claim of pointwise oracle dominance.

Table 4 examines center accuracy for three support sizes in one fixed configuration. Near the threshold, the selected center has visible shrinkage bias. Its deterministic error envelope covers that bias, which helps explain the conservative coverage. At a diffuse signal the known-direction oracle uses information that the selected interval does not exploit.

Support	Procedure	Bias (MCSE)	RMSE	Length	Coverage
1	Selected	-0.0555 (0.00025)	0.0566	0.962	1.0000
1	Direct	0.01 (0.0223)	0.997	3.92	0.9450
1	Oracle	0.000204 (0.00025)	0.0112	0.0433	0.9465
7	Selected	-0.352 (0.000346)	0.353	0.962	1.0000
7	Direct	-0.0272 (0.022)	0.985	3.92	0.9575
7	Oracle	0.000235 (9.21×10^{-5})	0.00412	0.0164	0.9550
8192	Selected	-0.011 (4.36×10^{-6})	0.011	0.962	1.0000
8192	Direct	0.00976 (0.0229)	1.02	3.92	0.9445
8192	Oracle	5.78×10^{-6} (2.74×10^{-6})	0.000122	0.000479	0.9480

Table 4: Center accuracy and length at $K = 3$, $p = n = 8192$, $a = M = \sigma = 1$ and cap one. Bias, its Monte Carlo standard error in parentheses, RMSE and length are divided by τ. Each row uses 2,000 replications. The support sizes represent concentrated, near-threshold and diffuse signals; the complete grid is retained in the results.

8.3 Weak rank and noise assumptions

For the boundary experiment, set $K = 3$ and cap two, and generate

$$A_\epsilon = \sqrt{1-\epsilon^2}\, U_1 + \epsilon U_2, \qquad \epsilon \in \{0, 0.001, 0.01, 0.1, 0.5\},$$

where U_1, U_2 are unit product tensors with disjoint mode supports. Each constrained unfolding has nonzero singular values $\sqrt{1-\epsilon^2}$ and ϵ for $\epsilon > 0$; the actual rank becomes one at zero. Figure 2 shows coverage and length for concentrated supports at $p = 8192$. The selected radius remains constant as ϵ vanishes. This demonstrates validity across the tested boundary and explicitly demonstrates the absence of actual-rank adaptation.

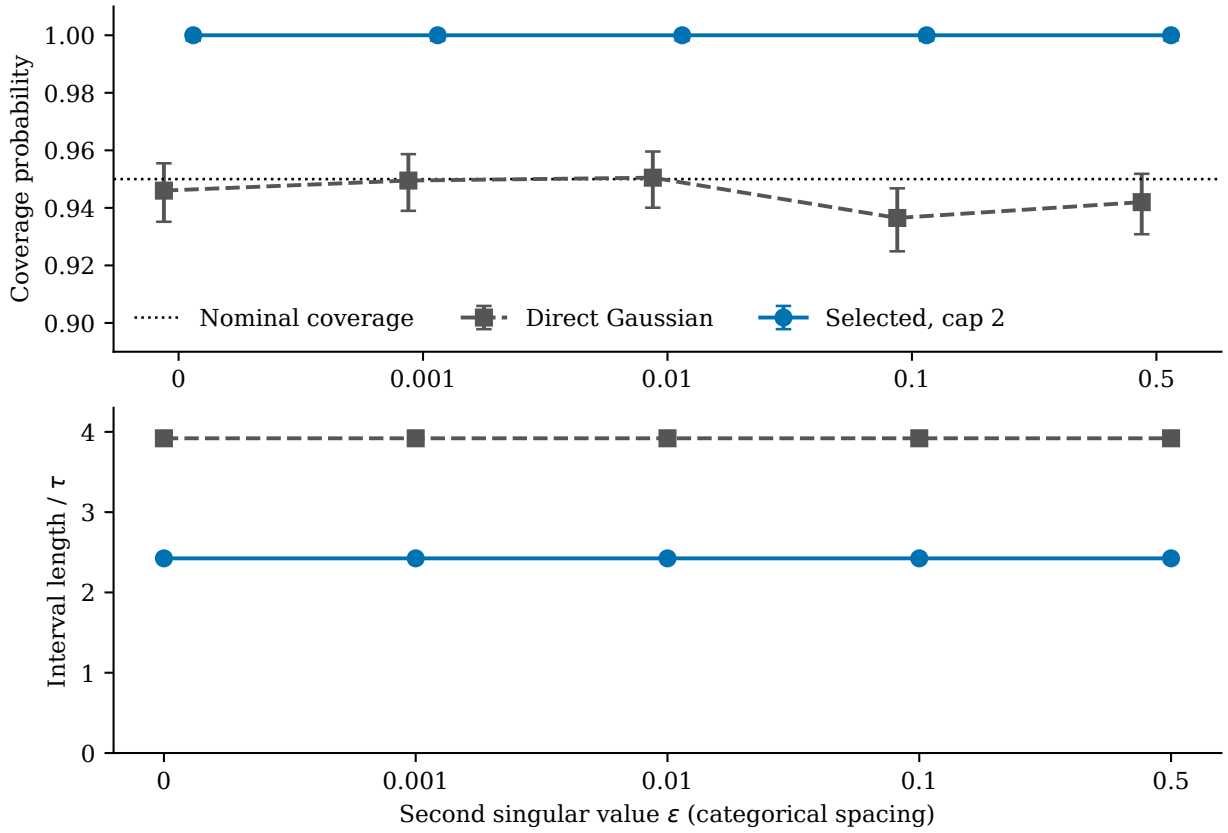


Figure 2: Coverage persists in the tested weak-component sequence, but the cap-based radius does not adapt to the actual rank. The setting is $K = 3$, $p = n = 8192$, $M = \sigma = 1$ and cap two, with disjoint single-coordinate product factors. Top: empirical coverage and exact 95% binomial intervals from 2,000 replications per value. Bottom: deterministic length divided by τ. The horizontal spacing in ϵ is categorical.

Three additional zero-mean stress configurations use independent variance-one Student t_3 errors, a Gaussian mixture with 5% of noise standard deviations multiplied by ten, and equicorrelated Gaussian coordinates with correlation 0.2. Table 5 reports the results. Variance contamination causes severe undercoverage for both procedures. Correlated Gaussian coordinates invalidate the direct iid variance but preserve the marginal Gaussian union bound used for the nonlinear interval. The Student result at zero is not a robustness theorem or evidence of coverage elsewhere.

Noise	Direct	Selected
gaussian	0.9540	1.0000
student t3	0.9585	0.9670
variance contamination	0.5740	0.3830
correlated	0.1430	1.0000

Table 5: Noise-assumption stress at $A = 0$, $K = 3$, $p = n = 512$, cap one and nominal level 0.95. Only the first row belongs to the stated iid Gaussian model. These results delimit the calibration; they do not support a robustness claim.

8.4 Limited scan implementation

A separate prototype implements complete support levels one and two with predetermined sphere nets. At level one the sphere is $\{-1, 1\}$; at level two an equally spaced circle net with $\lceil 2\pi K \rceil$ points has covering distance at most $1/(2K)$. Every product direction is evaluated, in batches. Gaussian thresholds use the actual cardinality and a simultaneous error budget.

For $K = 3$, $p \in \{8, 16, 32\}$, $\tau = 1$ and $M = \sqrt{p}$, we evaluated zero, one-coordinate and diffuse-support rank-one signals, with 100 replications per configuration and seed 20260912. Table 6 reports zero-signal runtimes; all nine configurations and Monte Carlo uncertainty are retained in the project. The original structural interval is selected at these dimensions: every observed length is 2τ and coverage is one in the 100 replications. The corresponding exact 95% binomial interval is approximately $[0.9638, 1]$. These are implementation and cost checks, not validation of minimax projection or evidence of a finite-dimensional precision improvement.

p	Product contractions	Median time (ms)
8	192,116	0.197
16	823,208	1.304
32	3,402,320	5.737

Table 6: Complete level-one and level-two scan costs for the prototype. Timing excludes noise generation and net preparation, uses batches and one OpenBLAS thread, and reflects the recorded macOS ARM64 environment.

8.5 New robust and adaptive experiments

Four separate suites address experiments absent from the preceding report. They use fixed seeds, exact binomial confidence intervals, per-configuration source/result records and timing measurements. There are 106 configurations and 76600 Monte Carlo repetitions. Earlier experiment outputs are unchanged. The new code implements the separated SVD/projection interval, an efficient threshold-based loading correction, directional medians on a complete small product-net scan, and approximate-rank threshold intervals. It does not implement the ideal minimax residual optimizer.

Actual-rank calibration and precision. At $K = 3, p = 8$, the mean is a sum of zero, one, or two equal diagonal product components. Nonzero singular values are 0.25, 3, 100, or 1000 times the explicit pilot bound. Two independent observations are generated under Gaussian, Rademacher, or correlated Gaussian errors; the latter have covariance $0.5I + 0.5uu^\top$ for a unit flat vector u. Loadings are the equal diagonal or its normalized mixture with a known off-diagonal product atom. Each of

54 configurations uses 500 repetitions. Forty-two configurations satisfy the separation condition; twelve deliberately violate it. The algorithm is supplied M and noise scales but no ranks.

Over the valid configurations, minimum observed coverage is 0.944, with exact 95% binomial interval $[0.9201, 0.9625]$, and all ranks are recovered in all repetitions. These extrema are descriptive and are not simultaneous coverage guarantees. For Gaussian equal loading at strength 1000, rank-one and rank-two mean length/direct-length ratios are 0.480 and 0.665, with coverages 0.992 and 0.988. At strengths 3 and 100 the bias certificate selects the direct interval. Thus identification occurs well before useful precision improvement under this conservative certificate. All twelve below-gap configurations have zero coverage because thresholded ranks are zero although the target is not. This explicitly rejects transferring the method's coverage to weak signals.

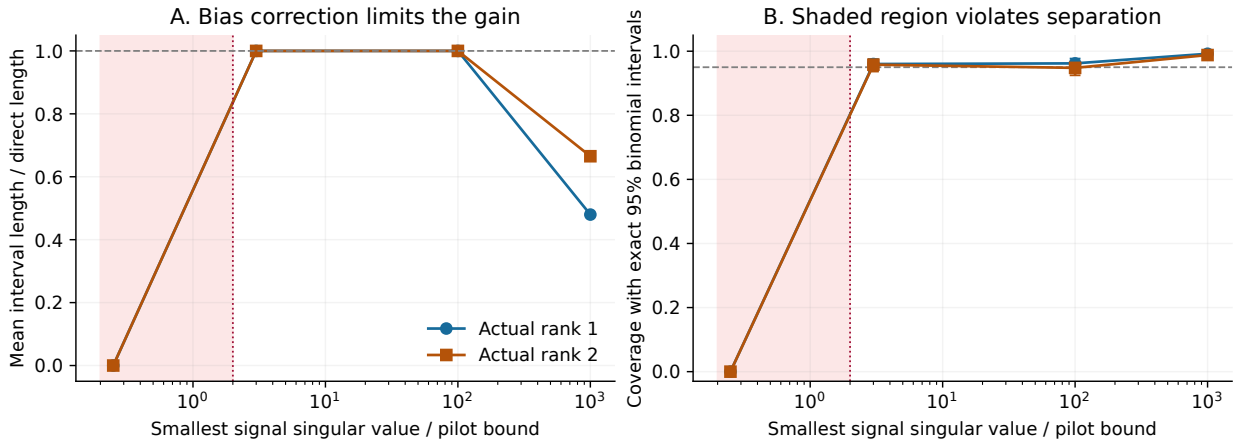


Figure 3: Implemented rank calibration under Gaussian noise and equal loading. Left: expected-length estimates relative to the direct interval. Right: coverage and exact binomial uncertainty. The shaded range violates the theorem's separation assumption. Large multiples of the conservative pilot bound represent extremely strong signals.

General loading correction. For $K = 4$, $p \in \{256, 4096\}$, $M = \sqrt{p}$ and noise scale one, use $L = \sqrt{1-\epsilon^2}L_{\text{flat}} + \epsilon U_{\text{off}}$ with $\epsilon \in \{0, 0.03, 0.3\}$. The off-diagonal unit product atom is orthogonal to the flat diagonal loading. Coordinate, diffuse rank-one and off-diagonal rank-one signals, with Gaussian or independent sign errors, produce 36 configurations with 1000 repetitions each. Only the required diagonal and off-diagonal observations are generated; these have exactly the same joint law as the contractions of the full tensor. The efficient interval combines the earlier threshold center with the measured residual loading term and the required noise radius, using deterministic structural/direct fallbacks. Coverage is at least 0.998 on this grid. At Gaussian $p = 4096$, the three remainder sizes yield lengths 0.794, 0.928, and 2.102, compared with direct length 3.920. Omitting the residual correction gives zero coverage for the off-diagonal signal at the two positive remainder sizes. These comparisons support correct calibration of the implemented upper construction, not minimax optimality for arbitrary loadings.

Finite-variance full scan. At $p = 2, K = 3$, all support levels are evaluated using the two singleton contractions and 19^3 product directions on the full block, for 6861 distinct evaluated directions. A circle net with 19 points has the required covering distance. Actual direction counts are used in the block calibration, which is valid in place of a larger cardinality upper bound. There

are $n = 128$ independent replicates and 400 repetitions under each of Gaussian, independent signs, variance-one Student t_3, and variance-one rare-outlier errors. The last law is $\sqrt{0.1}g + \sqrt{0.9/\varepsilon}\, B\xi$, with independent $g \sim N(0,1)$, $B \sim \text{Bernoulli}(\varepsilon)$, sign ξ, and $\varepsilon = 10^{-4}$, independently across entries and replicates.

The simultaneous median envelope covers in 400/400 repetitions for every law; the exact 95% lower confidence bound is approximately 0.991. Gaussian calibration of sample means covers in 399/400, 399/400, 390/400, and 349/400, respectively. Median thresholds are conservative. This tests the full implemented noise envelope, which is invariant to adding a mean, and does not simulate confidence-set extrema or an optimized minimax interval. Median runtime is approximately 1.6 milliseconds per repetition in the recorded single-thread environment, excluding noise generation and net construction.

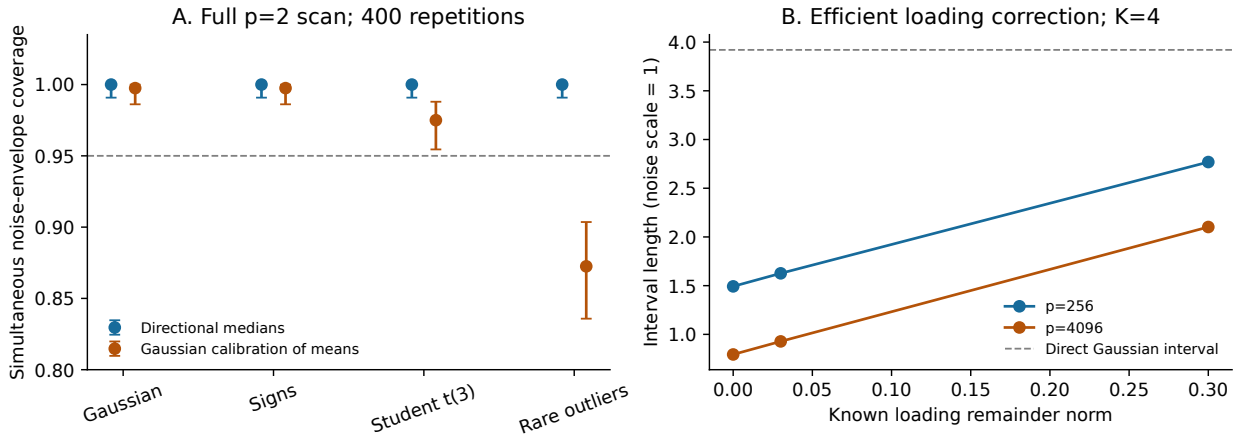


Figure 4: Left: simultaneous full-net envelope coverage with exact binomial intervals. Gaussian mean calibration is deliberately applied outside its proxy-one assumptions for the two heavy-tailed laws. Right: lengths of the efficient corrected-loading interval under Gaussian noise.

Approximate rank and independent checks. At $K = 4, p = 4096, M = 64$, a zero, coordinate or diffuse rank-one base is perturbed by ηL_{flat}, with $\eta \in \{0, 0.1, 0.5, 2\}$. Nonzero perturbations have full diagonal rank and norm η. The implemented threshold interval uses the previously proved approximation inflation, with deterministic direct/structural selection. Each configuration has 1000 repetitions; minimum observed coverage is 0.944. At a zero base and $\eta = 0.5$, inflated coverage is one and uninflated coverage is zero. At $\eta = 2$ the direct interval is selected. This tests a computable valid approximate-rank procedure, not the new sharp residual projection.

Separate finite mathematical checks pass for 200 projector/bias cases, 26595 median-tail comparisons, 112086 growing-rank support pairs and 100 shifted-block constructions. They include projection self-adjointness and idempotence, the pairwise second-order bound, integration of the median excess tail, and full likelihood mixture calculations. These checks supplement proofs rather than replace them. A certified implementation of the ideal robust optimizer, a tighter moderate-signal bias certificate, and a fair implementation comparison under a common regular-signal model remain outstanding.

9 Discussion

The results show that the parameter class over which coverage is required can determine the order of an honest interval's length. For a prescribed orthogonal contrast, simultaneous tensor rank restrictions yield an interval smaller than the order required by a single-unfolding relaxation. The comparison holds at arbitrarily weak signals and does not depend on identifying the actual rank. It also separates nonlinear procedures from every fixed-radius affine interval on the original class.

The exact diagonal reduction gives a precise connection with classical sequence inference. At rank one, the attainable diagonal vectors form an $\ell_{2/K}$ ball; Proposition 1 then identifies the optimal rate among diagonal-based intervals. The residual confidence-set projection and sparse harmonic mixture quantify the value of off-diagonal observations. At $M = \tau\sqrt{p}$, both $\mathcal{L}_0$ and $\mathcal{L}_{\max}$ have rate $\tau(\log p)^{3/2}/\sqrt{p}$ for $K = 3$, and $\tau\sqrt{\log p/p}$ for fixed $K \geq 4$. These are statistical minimax statements; the exact projection may be computationally expensive. Similarly, Theorem 4 applies to a larger honesty class, not to every algorithm that represents its data as a matrix.

Several restrictions affect practical inference. The loading must have the orthogonal product form, the Frobenius and rank caps are supplied, and the noise standard deviation is known. Appendix C improves the cap coefficient and relaxes the noise and exact-rank assumptions under explicit calibration. These refinements do not provide automatic adaptation when a singular value vanishes. The experiments show substantial conservatism even though length improves over the specified benchmarks. Strong-signal subspace inference or a known-direction oracle may give better pointwise precision.

Theorem 13 removes the original norm/noise scaling restriction, and Theorem 14 supplies squared-risk control even for arbitrarily large M/τ. Theorem 15 gives nonuniform orthogonal loading bounds, while Proposition 5 identifies sharp support-size dependence. These loading bounds need not match for arbitrary unequal weights. The remaining computational question is whether an efficient procedure can attain the full-data minimax interval rate. The truncated scan gives an explicit search budget, while Theorem 11 rules out minimax squared-error estimation by a range of polynomial degrees. This restricted result cannot be promoted to a lower bound against all polynomial-time interval procedures. Sharp general loading geometry and actual-rank adaptation remain further questions: they require a characterization of which contrasts benefit from joint constraints and which smaller-rank subclasses can be distinguished while retaining outer-class honesty.

Section 7 supplies positive results with precise boundaries. Full-data upper rates require directional concentration, not Gaussianity; finite-variance replicate rates use calibrated directional medians and sufficient replication. Minimax lower bounds over those unknown-law families follow from their Gaussian subfamilies, not from a universal equivalence of noise laws. Approximate ranks admit a sharp additive allowance for interval length. The separated SVD/projection construction is computable for arbitrary loadings and selects actual mode ranks, but its bias certificate can require very strong signals before length improves. Its coverage is not asserted on the original unseparated class. A convex outer-class obstruction limits full convexification, while the shifted-mixture bound extends the adaptation obstruction beyond zero. None of these statements supplies an efficient minimax interval in the original difficult regime.

Applications require a sampling model that justifies the noise calibration and an independently specified contrast. Repeated calibrated tensor measurements could provide such a model, but no real-data experiment is claimed here. In particular, arrays indexed by LLM model, layer and prompt generally have prompt, token and layer dependence, heterogeneous noise and data-derived loadings. Simply stacking those observations does not establish the Gaussian experiment. The noise-envelope extension in Appendix C is one ingredient, but a justified dependence model and functional-selection

theory would still be necessary before an LLM experiment could substantiate these inferential claims.

Reproducibility. The accompanying project contains the simulation protocol, modular Python code, fixed seeds, 817 result rows, exact binomial uncertainty, source and result hashes, the literature matrix and a theorem-status ledger. The original simulation outputs are reused without changing the interval algorithm. The theory review adds exact finite checks of the new sign-mixture calculation, approximation inequalities and affine bounds. The earlier review did not simulate the exhaustive scan. The subsequent development includes exact sparse-mixture and Hermite-moment checks and a finite-net prototype. All displayed numerical values come from saved outputs or analytic formulas. The norm/loading extension adds separate finite checks of the uniform-rate envelopes, weighted full-tensor mixtures, residual-noise tails, and orthogonal embeddings; these are not simulations of the ideal residual optimizer.

A Complete proofs

The proofs apply in finite dimensions. Constants used in the main text are explicit. All tensors and Gaussian variables are real.

A.1 Canonical product expansions

Lemma 1 (Controlled squared component weights)**.** *For a tensor in either cap class there is a finite representation*

$$A = \sum_{j=1}^{J_A} x_{1j} \otimes \cdots \otimes x_{Kj}, \qquad a_j^{\mathrm{cp}} = \prod_{k=1}^{K} \|x_{kj}\|_2,$$

such that $J_A \le J$ *and* $\sum_j (a_j^{\mathrm{cp}})^2 = \|A\|_F^2$.

Roadmap. Choose orthonormal Tucker mode bases or left-orthogonal TT cores; expand product terms and sum their squared weights. The TT argument uses a prefix identity, not orthogonality of individual paths.

Proof. The zero tensor has a zero-term representation. For a nonzero Tucker tensor, choose orthonormal bases U_k for the column spaces of its mode unfoldings. Orthogonal projection onto each such space leaves A unchanged. Contracting these bases gives a core $\mathcal{G}$ with $A = \mathcal{G} \times_1 U_1 \cdots \times_K U_K$ and $\|\mathcal{G}\|_F = \|A\|_F$. The actual mode ranks satisfy $r_k \le \bar{r}_k$.

Fix a mode k. Expand all core indices other than k. Each tuple j_{-k} gives unit factors $U_\ell[:, j_\ell]$ in modes $\ell \ne k$ and factor $U_k \mathcal{G}[:, j_{-k}]$ in mode k. Its squared product weight is $\|\mathcal{G}[:, j_{-k}]\|_2^2$. These slice squares sum to $\|\mathcal{G}\|_F^2$. The number of terms is $\prod_{\ell \ne k} r_\ell$. Choosing k to minimize the corresponding cap product proves the Tucker assertion.

For a TT tensor, sequential matrix factorizations yield cores $G_k \in \mathbb{R}^{r_{k-1} \times p_k \times r_k}$, with $r_0 = r_K = 1$ and $r_k \le \bar{r}_k$. At each cut, factor the relevant matrix through a basis of its column space. Equivalently, left-orthogonalize the cores by QR and absorb the triangular factors into the next core. Thus the $(r_{k-1} p_k) \times r_k$ matrix associated with G_k has orthonormal columns for $k < K$, and $\|G_K\|_F = \|A\|_F$. These steps use no inverse lower bound on a singular value.

Expand the contraction over all internal paths $(a_1, \dots, a_{K-1})$. A path contributes the product tensor with physical vectors $G_k[a_{k-1}, :, a_k]$. There are at most $\prod_{k<K} r_k \le J$ paths. To sum their squared weights, let

$$t_k(b) = \sum_{a_1,\dots,a_{k-1}} \prod_{\ell=1}^{k} \|G_\ell[a_{\ell-1}, :, a_\ell]\|_2^2, \qquad a_k = b.$$

Column normalization gives $t_1(b) = 1$. If $t_{k-1}(a) = 1$ for all a, then

$$t_k(b) = \sum_a t_{k-1}(a) \|G_k[a, :, b]\|_2^2 = 1$$

for $k < K$. Hence the sum of all squared path weights is

$$\sum_a t_{K-1}(a) \|G_K[a, :, 1]\|_2^2 = \|G_K\|_F^2 = \|A\|_F^2.$$

The individual path terms need not be orthogonal; the prefix identity is what proves the required equality. □

A.2 Proof of Theorem 1

Roadmap. Apply subadditivity, modewise Hölder and Bessel, then control the component-weight sum.

Proof. Let $q = 2/K \le 1$ and use Lemma 1. Subadditivity of the qth power of absolute value gives

$$\sum_i |\langle A, V_i \rangle|^q \le \sum_j \sum_i \prod_{k=1}^{K} |\langle x_{kj}, v_{ki} \rangle|^{2/K}.$$

Hölder's inequality with K equal exponents, followed by Bessel's inequality for each orthonormal family, yields

$$\begin{aligned} \sum_i \prod_k |\langle x_{kj}, v_{ki} \rangle|^{2/K} &\le \prod_k \left(\sum_i |\langle x_{kj}, v_{ki} \rangle|^2 \right)^{1/K} \\ &\le \prod_k \|x_{kj}\|_2^{2/K} = (a_j^{\mathrm{cp}})^q. \end{aligned}$$

Finally, the power-mean inequality (or Hölder with exponents $2/q$ and $2/(2-q)$) gives

$$\sum_j (a_j^{\mathrm{cp}})^q \le J_A^{1-q/2} \left\{ \sum_j (a_j^{\mathrm{cp}})^2 \right\}^{q/2} \le J^{1-q/2} M^q.$$

This is (14). For $J_A = 0$ both sides of the first inequality are zero. □

A.3 Exact rank-one diagonal image

Lemma 2 (Exact diagonal reduction)**.** *Under (4), the image of the rank-one tensor Frobenius ball of radius M under $A \mapsto (\langle A, V_i \rangle)_{i=1}^m$ is exactly*

$$\left\{ a \in \mathbb{R}^m : \sum_i |a_i|^{2/K} \le M^{2/K} \right\}.$$

Roadmap. Use the diagonal bound for one inclusion and construct signed product factors for the converse.

Proof. One inclusion follows from Theorem 1 with $J = 1$. Conversely, for any vector in the displayed ball, put

$$x_1 = \sum_i \operatorname{sign}(a_i)|a_i|^{1/K} v_{1i}, \qquad x_k = \sum_i |a_i|^{1/K} v_{ki}, \quad k \geq 2,$$

and set $A = x_1 \otimes \cdots \otimes x_K$. Orthonormality gives $\langle A, V_i \rangle = a_i$ and

$$\|A\|_F = \prod_k \|x_k\|_2 = \left(\sum_i |a_i|^{2/K} \right)^{K/2} \leq M.$$

The tensor has every Tucker and TT rank at most one. The zero vector is handled by the zero tensor. □

A.4 Threshold error and structural radius

Lemma 3 (Scalar threshold envelope)**.** *If $|e| \leq \lambda$ and $s = \operatorname{soft}_\lambda(x + e)$, then s lies between zero and x and*

$$|s - x| \leq \min(|x|, 2\lambda).$$

For $q \in (0, 1]$, $t, u \geq 0$, one also has $\min(t, u) \leq t^q u^{1-q}$, with the continuous interpretation at zero.

Roadmap. Check positive and negative scalar inputs, then interpolate the two deterministic error bounds.

Proof. For $x \geq 0$, $x + e \geq -\lambda$, so thresholding cannot produce a negative value. Also $x + e - \lambda \leq x$, so $0 \leq s \leq x$. If $x > 2\lambda$, then $x + e > \lambda$ and $x - s = \lambda - e \leq 2\lambda$. Otherwise $x - s \leq x \leq 2\lambda$. The negative case follows by oddness. If $t \leq u$, then $t = t^q t^{1-q} \leq t^q u^{1-q}$; if $u \leq t$, then $u = u^q u^{1-q} \leq t^q u^{1-q}$. □

Lemma 4 (Matrix functional bound)**.** *If $\operatorname{rank}(B) \leq r$ and H has singular values $s_1(H) \geq s_2(H) \geq \cdots$, then*

$$|\langle H, B \rangle| \leq \|B\|_F \left(\sum_{i \leq r} s_i(H)^2 \right)^{1/2}.$$

Roadmap. Project onto the matrix column space and maximize the spectral trace over projections of bounded rank.

Proof. Let P project onto the column space of B. Then $PB = B$ and Cauchy–Schwarz gives $|\langle H, B \rangle| = |\langle PH, B \rangle| \leq \|PH\|_F \|B\|_F$. Write $HH^\top = \sum_i \lambda_i u_i u_i^\top$, with decreasing nonnegative eigenvalues. The numbers $b_i = u_i^\top P u_i$ satisfy $0 \leq b_i \leq 1$ and $\sum_i b_i = \operatorname{tr}(P) \leq r$. Therefore $\|PH\|_F^2 = \sum_i \lambda_i b_i \leq \sum_{i \leq r} \lambda_i$: moving any weight from a smaller eigenvalue to an unfilled larger one cannot reduce this sum. Since $\lambda_i = s_i(H)^2$, the result follows. □

A.5 Proof of Theorem 2

Roadmap. Construct a uniform coordinate-noise event, bound the threshold error pathwise, verify the other intervals and use the deterministic selection rule.

Proof. The product atoms are orthonormal because

$$\langle V_i, V_j \rangle = \prod_k \langle v_{ki}, v_{kj} \rangle = \mathbf{1}\{i = j\}.$$

Gaussian isotropy then gives the independent normals in (5). For a standard normal g, Chernoff's inequality gives $\mathbb{P}(g > t) \le \exp(-t^2/2)$ by minimizing $\exp(-ut + u^2/2)$ at $u = t$. Apply the same inequality to $-g$ and take a union bound to obtain

$$\mathbb{P}\{\max_i |\tau g_i| > \lambda\} \le 2m \exp\{-\lambda^2/(2\tau^2)\} = \alpha.$$

On the complementary event, Lemma 3 and Theorem 1 give

$$\begin{aligned} |\widehat{\psi}_{\mathrm{NL}} - \psi(A)| &\le \|w\|_\infty \sum_i \min(|a_i|, 2\lambda) \\ &\le \|w\|_\infty (2\lambda)^{1-q} \sum_i |a_i|^q \le h_{\mathrm{NL}}. \end{aligned}$$

This proves the uniform coverage of I_{NL}.

Because $\|L\|_F = 1$, $\langle L, Z \rangle - \psi(A) \sim N(0, \tau^2)$, so I_G has exact coverage. For any constrained unfolding, the loading matrix has singular values $|w_i|$: the product vectors on each side form orthonormal families. Lemma 4 thus gives $|\psi(A)| \le M(\sum_{i \le r} w_{(i)}^2)^{1/2}$. Choosing the smallest cap proves $|\psi(A)| \le h_0$, so I_0 has coverage one. The selected index in (13) is deterministic, hence the selected interval retains its individual coverage without a multiple-selection correction. Its length is exactly (15). □

A.6 Proof of Theorem 3

Roadmap. Force a variance cost at zero and a bias cost at signed coordinate tensors, then combine the two necessary inequalities.

Proof. Set $t = \langle W, L \rangle$. At $A = 0$ the center error is normal with mean b and standard deviation $v = \tau \|W\|_F$. For fixed h and $v > 0$, the probability that this variable belongs to $[-h, h]$ is maximized at $b = 0$. Indeed, for $b \ge 0$ its derivative is

$$v^{-1}\{\phi((h + b)/v) - \phi((h - b)/v)\} \le 0,$$

since $|h+b| \ge |h-b|$; the function is even in b. Honesty at zero therefore implies $h \ge z\tau \|W\|_F \ge z\tau |t|$. If $v = 0$, the same last inequality is automatic.

At $A = \pm M E_i$, the mean center error is

$$b \pm M(\langle W, E_i \rangle - p^{-1/2}).$$

At least one sign has absolute mean at least $M|\langle W, E_i \rangle - p^{-1/2}|$. A symmetric interval whose radius is strictly smaller than this mean magnitude has probability below $1/2$ under a nondegenerate

normal distribution: it lies strictly to one side of the median. In the degenerate case the coverage is zero. Since $1-\alpha > 1/2$, honesty forces

$$h \geq M \max_i |\langle W, E_i\rangle - p^{-1/2}| \geq \frac{M}{\sqrt{p}}|1-t|.$$

The last inequality follows from $t - 1 = p^{-1/2}\sum_i(\langle W, E_i\rangle - p^{-1/2})$. Put $a = M/\sqrt{p}$ and $b' = z\tau$. The two bounds imply

$$1 \leq |1-t| + |t| \leq h/a + h/b',$$

which rearranges to (17). On the rank-one tensor ball, $h_0 = M/\sqrt{p}$, and the direct/structural choice gives the matching order. No result about arbitrary data-dependent widths is invoked. □

A.7 A mixture inequality for confidence intervals

Lemma 5 (Common-target alternatives)**.** *Let P_0 have target zero and let Q be a mixture of alternatives whose targets all equal $\delta > 0$. If C covers zero under P_0 and covers the target at every alternative with probability at least $1-\alpha$, then*

$$\mathbb{E}_0|C| \geq \delta\{1 - 2\alpha - \mathrm{TV}(P_0, Q)\}. \tag{48}$$

If $Q \ll P_0$, then $\mathrm{TV}(P_0, Q) \leq \frac{1}{2}\sqrt{\chi^2(Q, P_0)}$.

Roadmap. Transfer coverage from the common-target mixture to the null using total variation, then intersect the two coverage events.

Proof. Mixture coverage gives $Q(\delta \in C) \geq 1-\alpha$. By the definition of total variation, $P_0(\delta \in C) \geq 1 - \alpha - \mathrm{TV}(P_0, Q)$. Intersecting with $\{0 \in C\}$ gives probability at least $1 - 2\alpha - \mathrm{TV}$. An interval containing both points has length at least δ, proving (48). The divergence bound follows from Cauchy–Schwarz applied to $\frac{1}{2}\mathbb{E}_0|dQ/dP_0 - 1|$. Appending an independent random seed leaves total variation unchanged, so randomized intervals are covered as well. □

For Gaussian shifts $P_B = N(\mathrm{vec}(B), \tau^2 I_D)$, direct integration of their likelihood ratios gives

$$\mathbb{E}_0\left[\frac{dP_B}{dP_0}\frac{dP_C}{dP_0}\right] = \exp\{\langle B, C\rangle/\tau^2\}. \tag{49}$$

To verify this, each ratio is $\exp\{\langle Z, B\rangle/\tau^2 - \|B\|_F^2/(2\tau^2)\}$. Multiply the ratios and use the moment generating function of $\langle Z, B+C\rangle$, whose variance under P_0 is $\tau^2\|B+C\|_F^2$. The exponent simplifies to (49).

A.8 Proof of Theorem 4

Roadmap. Construct rank-one matrix alternatives within the chosen unfolding and bound their full Gaussian likelihood second moment using an auxiliary Gaussian.

Proof. For the given bipartition, let $f_i = \bigotimes_{k\in S} e_i$ and $g_i = \bigotimes_{k\notin S} e_i$. Both families are orthonormal. Let $u = p^{-1/2}(s_1, \ldots, s_p)$ with independent uniform signs, and define

$$\theta = \min(M, \tau\sqrt{p}/4), \qquad A_u = \theta\left(\sum_i u_i f_i\right) \otimes \left(\sum_j u_j g_j\right).$$

Every alternative has rank one in this unfolding, norm θ, and functional value $\delta = \theta/\sqrt{p}$. Let Q average their Gaussian laws. If v is an independent copy of u, then (49) gives

$$1 + \chi^2(Q, P_0) = \mathbb{E}_{u,v} \exp\{\theta^2 (u^\top v)^2/\tau^2\} = \mathbb{E} \exp\{a S_p^2/p\},$$

where S_p is a sum of p independent signs and $a = \theta^2/(\tau^2 p) \le 1/16$.

Introduce an independent standard normal H. Its moment generating function yields

$$\begin{aligned}\mathbb{E} \exp(a S_p^2/p) &= \mathbb{E}_H \left\{ \cosh(\sqrt{2a/p}\, H) \right\}^p \\ &\le \mathbb{E}_H \exp(a H^2) = (1-2a)^{-1/2} \le \sqrt{8/7}.\end{aligned}$$

The inequality uses $\log\cosh(x) \le x^2/2$: for $x \ge 0$, its derivative $\tanh(x)$ is at most x, and both functions are even. The Gaussian-square expectation follows by integrating $\exp\{-(1-2a)h^2/2\}/\sqrt{2\pi}$; it is finite since $a < 1/2$. Nonnegative integrands justify exchanging the expectations.

Lemma 5 now gives (19). The constant is positive for $\alpha < 1/4$. On Θ_S, Lemma 4 with loading singular values all equal to $p^{-1/2}$ gives $|\psi(A)| \le M/\sqrt{p}$. Choosing between this structural interval and the direct interval attains the lower order. All calculations use the entire Gaussian experiment; the result does not arise from hiding off-diagonal observations. □

A.9 Proof of Theorem 5

Roadmap. Substitute the specified dimensions and norm scaling, then apply the two lower-bound theorems with their confidence-level restrictions.

Proof. Substitute $M = \tau\sqrt{p}$, $q = 2/K$ and $\|w\|_\infty = p^{-1/2}$ into (10). This gives

$$h_{\mathrm{NL}} = \tau J^{1-1/K} p^{1/K - 1/2} \{8 \log(2p/\alpha)\}^{(K-2)/(2K)},$$

which is the radius in (20). For fixed $K \ge 3$, its exponent $\beta = (K-2)/(2K)$ is positive and $(\log p/p)^\beta \to 0$. The cap classes contain all $\pm M E_i$ and zero, so Theorem 3 gives radius at least $z\tau/(1+z)$. For any retained constrained unfolding, the corresponding matrix-rank cap class contains Θ_S, and Theorem 4 gives expected length at zero at least $c_\alpha \tau/4$. The same comparisons under the stated growing-cap condition follow by retaining the factor $J^{1-1/K}$. □

A.10 Proof of Theorem 6

Roadmap. Mix orthogonal coordinate shifts, compute the full likelihood second moment exactly and apply the interval mixture lemma.

Proof. Let $a = \min\{M, \tau\sqrt{\log(1+p/4)}\}$ and mix uniformly over $A_j = aE_j$, $j = 1, \dots, p$. Each is in the rank-one tensor class and has target $\delta = a/\sqrt{p}$. Orthogonality of the E_j and (49) imply

$$1 + \chi^2(Q, P_0) = p^{-2}\{p \exp(a^2/\tau^2) + p(p-1)\}.$$

Thus $\chi^2(Q, P_0) = \{\exp(a^2/\tau^2) - 1\}/p \le 1/4$ and $\mathrm{TV}(P_0, Q) \le 1/4$. Lemma 5 gives (21). This construction does not establish that these alternatives are least favorable. □

A.11 Proof of Proposition 1

Roadmap. Construct sparse diagonals inside the rank-one tensor class, bound the diagonal likelihood mixture using factorial moments and balance the support size. The full-data likelihood is kept separate.

Proof. For an integer $1 \le s \le p$, let S be a uniformly chosen size-s subset of $\{1, \dots, p\}$. Define

$$A_S = b_s \left(\sum_{i \in S} e_i \right)^{\otimes K}.$$

This is a rank-one tensor, $\|A_S\|_F = b_s s^{K/2} \le M$, and every alternative has target $\delta_s = s b_s / \sqrt{p}$. Its observed diagonal has mean $b_s \mathbf{1}_S$. Let Q_d be the uniform mixture of these diagonal Gaussian laws, and let $P_{0,d}$ be the zero-mean diagonal law. Applying the Gaussian shift calculation (49) in dimension p gives

$$1 + \chi^2(Q_d, P_{0,d}) = \mathbb{E}_{S,T} \exp\{b_s^2 |S \cap T| / \tau^2\}, \tag{50}$$

where S, T are independent uniform size-s subsets. This is the sparse-mixture mechanism used in classical sequence lower bounds [Collier et al., 2017]; the following counting argument records the needed bound explicitly.

Conditional on S, $H = |S \cap T|$ is hypergeometric with population size p, success count s and sample size s. Write $(x)_j = x(x-1)\cdots(x-j+1)$, with $(x)_0 = 1$. Counting the j-element subsets of S contained in T yields

$$\mathbb{E}\binom{H}{j} = \binom{s}{j} \frac{(s)_j}{(p)_j} \le \binom{s}{j} (s/p)^j, \qquad 0 \le j \le s.$$

For $j \ge 1$, the inequality follows factor by factor from $(s-\ell)/(p-\ell) \le s/p$ for $0 \le \ell < j$, since $s \le p$. Set $t = \exp(b_s^2/\tau^2) - 1 \ge 0$. Expanding $(1+t)^H$ and using the preceding factorial moments gives

$$\begin{aligned} \mathbb{E} \exp(b_s^2 H / \tau^2) &= \sum_{j=0}^{s} t^j \mathbb{E}\binom{H}{j} \\ &\le (1 + st/p)^s \le \exp(s^2 t / p) \le e^{1/4}. \end{aligned}$$

The last inequality uses the definition of b_s. Equation (50) and Cauchy–Schwarz therefore give

$$\mathrm{TV}(Q_d, P_{0,d}) \le \frac{1}{2} \sqrt{e^{1/4} - 1}.$$

Lemma 5, now applied to the diagonal experiment, proves (23). The constant is positive because $\alpha < 1/4$ and $\frac{1}{2}\sqrt{e^{1/4}-1} < 1/2$. The argument also covers independent randomization.

To prove (24), let $x_p = (p/\log p)^{1/K}$ and $s_p = \lfloor x_p \rfloor$. For sufficiently large p, $x_p/2 \le s_p \le x_p$ and $s_p \ge 1$. When $M = \tau\sqrt{p}$, the norm term in b_{s_p} is at least $\tau\sqrt{\log p}$. The noise term is at most this quantity, since $1 + p/(4s_p^2) \le 1 + p/4 \le p$ for $p \ge 2$. Thus the minimum equals the noise term. Since $K > 2$ and $s_p \le p^{1/K}$ eventually,

$$\log\{1 + p/(4s_p^2)\} \ge \log\{1 + p^{1-2/K}/4\} \ge \tfrac{1}{2}(1 - 2/K) \log p$$

for sufficiently large p. Consequently,

$$\frac{s_p b_{s_p}}{\sqrt{p}} \geq \frac{1}{2}\sqrt{\frac{1-2/K}{2}}\,\tau\left(\frac{\log p}{p}\right)^{(K-2)/(2K)}.$$

Theorem 2 with $J = 1$ gives an upper bound of the same order and deterministic length. Since the worst-case criterion is at least the criterion at zero, both rates follow. The fixed-cap extension follows by class inclusion for the lower bound and a constant $J^{1-1/K}$ for the upper bound.

For completeness, the corresponding *full-tensor* mixture would have $\langle A_S, A_T\rangle = b_s^2|S\cap T|^K$ in (49). Its likelihood second moment is not (50), and the preceding bound does not apply to it. The proposition is restricted to diagonal observations throughout. □

B Proofs of the full-data refinements

Lemma 6 (Finite product nets and injective norm)**.** *The nets used in* (25) *exist and satisfy* $\rho(B) \leq 2N_S(B)$ *for any tensor on* S^K*. Furthermore, a tensor* B *in either rank-cap class satisfies* $\|B\|_F \leq \sqrt{J}\,\rho(B)$*.*

Roadmap. Use a volume bound and product telescoping, then apply the canonical squared-weight identity.

Proof. A maximal ε-separated subset of the unit sphere is an ε-net. Balls of radius $\varepsilon/2$ around its points are disjoint and contained in the ball of radius $1+\varepsilon/2$. The ratio of their Euclidean volumes bounds the cardinality by $(1+2/\varepsilon)^s$. Compactness gives a finite maximal set. Take $\varepsilon = 1/(2K)$.

Approximate each factor of any unit product direction by a net point. Telescoping the product one factor at a time, each contraction difference has magnitude at most $\varepsilon\rho(B)$, by multilinearity and the definition of ρ. The sum is at most $K\varepsilon\rho(B) = \rho(B)/2$. Taking the supremum gives $\rho(B) \leq N_S(B) + \rho(B)/2$.

For the second assertion use Lemma 1, writing its product terms as X_j. Then

$$\|B\|_F^2 = \sum_j \langle B, X_j\rangle \leq \rho(B)\sum_j \|X_j\|_F \leq \rho(B)\sqrt{J}\left(\sum_j \|X_j\|_F^2\right)^{1/2} = \rho(B)\sqrt{J}\,\|B\|_F.$$

The middle inequality is Cauchy–Schwarz, and the last identity is the squared-weight conclusion of Lemma 1. Divide when $B \neq 0$; the zero case is immediate. □

B.1 Proof of Theorem 7

Roadmap. A finite Gaussian union bound controls every net contraction. Test acceptance on that event bounds each ordered diagonal coordinate. Summation gives the interval and intersection gives the two expected-length guarantees.

Proof. There are at most $(ep/s)^s$ subsets of size s, since $\binom{p}{s} \leq p^s/s!$ and $\log(s!) \geq \int_1^s \log x\,dx \geq s\log s - s$. Each unit product contraction of τG has distribution $N(0,\tau^2)$. Therefore the Gaussian Chernoff bound gives

$$\begin{aligned}\mathbb{P}\{\exists S : N_S(\tau G) > \lambda^{\mathrm{scan}}_{|S|}\} &\leq \sum_{s=1}^{p} 2(ep/s)^s(1+4K)^{Ks}\exp\{-(\lambda^{\mathrm{scan}}_s)^2/(2\tau^2)\}\\ &\leq \gamma.\end{aligned}$$

Let $\mathcal{E}_\gamma$ be the complementary event. No independence between the scanned contractions is required.

On $\mathcal{E}_\gamma$, if the test accepts, the triangle inequality for the finite maximum implies $N_S(A) \le 2\lambda_s^{\mathrm{scan}}$. Lemma 6 gives $\rho(A_{S,\dots,S}) \le 4\lambda_s^{\mathrm{scan}}$. Coordinate restriction cannot increase any constrained unfolding rank, because it multiplies an unfolding by selection matrices. Theorem 1 and Lemma 6 thus imply

$$\sum_{i\in S} |A_{i,\dots,i}|^q \le J^{1-q/2}\|A_{S,\dots,S}\|_F^q \le J\,\rho(A_{S,\dots,S})^q.$$

Let $|a|_{(s)}$ be the sth largest absolute diagonal entry. Choose S to contain the s largest entries. It follows that

$$s|a|_{(s)}^q \le J(4\lambda_s^{\mathrm{scan}})^q, \qquad |a|_{(s)} \le 4J^{K/2}\lambda_s^{\mathrm{scan}} s^{-K/2}.$$

Summing these inequalities and using $|\psi(A)| \le p^{-1/2}\sum_i |a_i|$ proves the bound $|\psi(A)| \le R_\gamma$ in (26). On rejection the structural interval covers surely. Therefore C_{scan} covers on $\mathcal{E}_\gamma$ and is honest at level $1-\gamma$.

Both intersected intervals cover with probability at least $1-\alpha_0-\gamma$. On that event the intersection is nonempty and covers; replacing empty intersections by $\{0\}$ preserves this argument. Its length is always at most ℓ_*. At zero, the rejection probability is at most γ, and on acceptance the intersection has length at most $2R_\gamma$. This gives (27), without independence between the intervals.

For the specified γ, elementary logarithm bounds give $\lambda_s^{\mathrm{scan}} \le C_K\tau\sqrt{s\log(2p/\alpha)}$. For example, use $s\log(ep/s) \le s(1+\log p)$, $\log(2p/\gamma) \le 2\log(2p/\alpha)$ and $s \ge 1$. Equation (26) now proves (28); the extra term $\ell_*\gamma$ is smaller in order by Theorem 5. If $K \ge 4$, the sum is at most $1 + 2/(K-3)$ by integration of $x^{-(K-1)/2}$ over $[1,\infty)$. If $K = 3$, it is at most $1+\log p$. Combining the upper bound for $K \ge 4$ with Theorem 6 proves (29). □

Lemma 7 (Uniform cubic moment for a weighted sign sum)**.** *For independent uniform signs ϵ_i, let $S_p = \sum_{i=1}^p i^{-2/3}\epsilon_i$. Then, uniformly in $p \ge 1$,*

$$\mathbb{P}(|S_p| > x) \le 2e^{-x^3/192} \quad (x \ge 12), \qquad \mathbb{E}\exp\{|S_p|^3/40000\} \le 20/19 + 1/100 < 5/4.$$

Roadmap. Split the weighted sign sum into a bounded head and a sub-Gaussian tail, then integrate its cubic tail probability.

Proof. For $x \ge 12$, put $m = \lfloor (x/6)^3 \rfloor$. If $m \ge p$, then $|S_p| \le 3p^{1/3} \le x/2$. Otherwise, the first m terms have absolute sum at most $3m^{1/3} \le x/2$. The remaining independent sign sum has variance proxy

$$V_m = \sum_{i>m} i^{-4/3} \le 3m^{-1/3} \le 24/x.$$

Here $m \ge (x/6)^3/2$ and $18\cdot 2^{1/3} < 24$. The product inequality $\prod_i \cosh(ti^{-2/3}) \le \exp(t^2V_m/2)$ and Chernoff optimization give a two-sided tail bound $2\exp\{-(x/2)^2/(2V_m)\} \le 2e^{-x^3/192}$.

Put $u = 1/40000$ and $Y = |S_p|^3$. Tonelli's theorem applied to $e^{uY} = 1 + \int_0^Y ue^{ut}\,dt$ yields

$$\mathbb{E}e^{uY} \le e^{1728u} + \frac{2u}{1/192-u}e^{-(1/192-u)1728} \le \frac{20}{19} + \frac{1}{100}.$$

The last step uses $1728u < 1/20$, $e^x \le (1-x)^{-1}$ for $0 \le x < 1$, and $1/192 - u \ge 1/200$. The exponential factor in the second term is at most one. These estimates are uniform in p. □

B.2 Proof of Theorem 8

Roadmap. Couple three rank-one factor signs so that the diagonal is deterministic and harmonic. Control the full likelihood mixture by the cubic moment lemma and Hölder, without assuming independence of the three inner-product sums.

Proof. For independent uniform sign vectors $u, v \in \{-1, 1\}^p$, define

$$(x_u)_i = u_i i^{-1/3}, \qquad A_{u,v} = \frac{\tau}{200} x_u \otimes x_v \otimes x_{uv},$$

where $(uv)_i = u_i v_i$. Every alternative is rank one and

$$\|A_{u,v}\|_F = \frac{\tau}{200} \left(\sum_i i^{-2/3} \right)^{3/2} \le \frac{3^{3/2}}{200} \tau \sqrt{p} \le M.$$

The sum bound follows from $1 + \int_1^p x^{-2/3}\, dx \le 3p^{1/3}$. Its diagonal is $\tau/(200i)$, so the common target is $\delta = \tau H_p/(200\sqrt{p})$.

Let Q mix the full Gaussian laws over u, v. For independent copies u', v', set $\epsilon_i = u_i u_i'$ and $\eta_i = v_i v_i'$. The full tensor inner product divided by τ^2 is $S_1 S_2 S_3/40000$, where

$$S_1 = \sum_i i^{-2/3} \epsilon_i, \quad S_2 = \sum_i i^{-2/3} \eta_i, \quad S_3 = \sum_i i^{-2/3} \epsilon_i \eta_i.$$

Each has the marginal distribution in Lemma 7. They are dependent. The likelihood identity (49), the arithmetic–geometric mean inequality and Hölder give, with $u_0 = 1/40000$,

$$\begin{aligned} 1 + \chi^2(Q, P_0) &= \mathbb{E} e^{u_0 S_1 S_2 S_3} \\ &\le \mathbb{E} e^{u_0 (|S_1|^3 + |S_2|^3 + |S_3|^3)/3} \\ &\le \prod_{j=1}^{3} \{ \mathbb{E} e^{u_0 |S_j|^3} \}^{1/3} < 5/4. \end{aligned}$$

Thus $\mathrm{TV}(Q, P_0) < 1/4$. Lemma 5 gives $\mathbb{E}_0 |C| \ge (3/4 - 2\alpha)\delta$, proving (30). Since $H_p \asymp \log p$, this also proves the claimed obstruction to the earlier zero-point conjecture. □

C Supporting refinements

These separately calibrated extensions do not change the observation model of a lower bound by implication. The simulations retain the original Gaussian procedure.

C.1 Diagonal geometry

Proposition 10 (Smaller rank coefficient)**.** *In $\sum_i |a_i|^{2/K} \le D M^{2/K}$ one may take*

$$D_{\mathrm{Tu}} = \min \left\{ J_{\mathrm{Tu}}^{1-1/K}, m^{1-1/K}, (\prod_k \bar{r}_k)^{1/K} \right\}.$$

For TT caps put $\bar{r}_0 = \bar{r}_K = 1$ and $\rho_k = \min(p_k, \bar{r}_{k-1} \bar{r}_k)$; one may take

$$D_{\mathrm{TT}} = \min \left\{ J_{\mathrm{TT}}^{1-1/K}, m^{1-1/K}, (\prod_k \rho_k)^{1/K} \right\}.$$

The nonlinear radius remains valid with coefficient D.

Roadmap. Add ambient and mode-subspace bounds to the original expansion bound.

Proof. Bessel gives $\sum_i |a_i|^2 \le M^2$; Hölder in dimension m gives the second coefficient. For Tucker mode projectors P_k put $\ell_{ki} = \|P_k v_{ki}\|_2^2$. Projection leaves A unchanged, so Cauchy–Schwarz and Hölder give

$$\sum_i |a_i|^{2/K} \le M^{2/K} \sum_i \prod_k \ell_{ki}^{1/K} \le M^{2/K} \prod_k (\sum_i \ell_{ki})^{1/K} \le M^{2/K} \prod_k \bar{r}_k^{1/K}.$$

The last step uses $\sum_i v_{ki}^\top P_k v_{ki} \le \operatorname{tr} P_k$. For TT cores, fixing the two bond indices of core k expresses its physical fibers through at most $\bar{r}_{k-1}\bar{r}_k$ vectors. The mode rank is therefore at most ρ_k; the Tucker argument applies. Theorem 1 gives the first coefficient. □

For equal Tucker caps r this supplies coefficient r, instead of $r^{(K-1)^2/K}$; optimal rank dependence is not established. For unit loading vectors with frame bounds $\sum_i v_{ki} v_{ki}^\top \preceq F_k I$, the original proof instead gives coefficient $J^{1-1/K} \prod_k F_k^{1/K}$. This follows by replacing each Bessel inequality by its frame bound. The direct variance must use the actual $\|L\|_F^2$; the orthogonal structural formula is not asserted for these frames.

C.2 Exact affine benchmark

Let $Q_\alpha(\mu, v)$ be the smallest $h \ge 0$ such that $\mathbb{P}\{|N(\mu, v^2)| \le h\} \ge 1 - \alpha$, for $\mu, v \ge 0$, with $Q_\alpha(\mu, 0) = \mu$.

Proposition 11 (Structural range and fixed affine optimum)**.** *The range of ψ on either cap class under (4) is exactly $[-h_0, h_0]$. Under (16) and $0 < \alpha < 1/2$, set $a_0 = M\sqrt{s_{\mathrm{cap}}/p}$. Among all honest full-data fixed-radius affine intervals, the optimal radius is*

$$h^*_{\mathrm{aff}} = \min_{0 \le t \le 1} Q_\alpha(a_0(1-t), \tau t).$$

Center $t\langle L, Z\rangle$ attains the optimum. If $a_0 \le z_{1-\alpha}\tau$, then $h^*_{\mathrm{aff}} = a_0$.

Roadmap. Attain the structural bound, force bias and variance for any affine center, and reduce to a scalar optimization.

Proof. Let T index the s_{cap} largest weights. The tensor $M \sum_{i\in T} w_i V_i / (\sum_{i\in T} w_i^2)^{1/2}$ has norm M, all constrained ranks at most s_{cap}, and target h_0. Its signed scalar multiples attain the full interval; Lemma 4 gives the reverse inclusion.

For center $\langle W, Z\rangle + b$ put $t = \langle W, L\rangle$ and $g_i = \langle W, E_i\rangle - p^{-1/2}$. All s_{cap}-sparse diagonal tensors of norm at most M belong to the class. Opposite such tensors force an absolute mean error at least

$$|b| + M(\sum_{i=1}^{s_{\mathrm{cap}}} g_{(i)}^2)^{1/2} \ge a_0|1 - t|.$$

Indeed, the top-square sum is at least s_{cap}/p times the full sum, and $\|g\|_2 \ge |\sum_i g_i|/\sqrt{p} = |t - 1|$. The noise standard deviation is $\tau\|W\|_F \ge \tau|t|$. Confidence greater than one half forces h to be at least the absolute mean error, by the normal median argument. For $h \ge \mu \ge 0$ the probability $F(h; \mu, v) = \Phi((h-\mu)/v) + \Phi((h+\mu)/v) - 1$ decreases with μ and with $v > 0$. For μ, compare the normal densities. For v, its derivative is

$$-\{(h-\mu)\phi((h-\mu)/v) + (h+\mu)\phi((h+\mu)/v)\}/v^2 \le 0.$$

Thus $h \geq Q_\alpha(a_0|1-t|, \tau|t|)$; zero variance follows directly. For $t < 0$ this is at least a_0, and for $t > 1$ it is at least $z_{1-\alpha/2}\tau$. Both endpoints are achievable. On $[0,1]$, center $t\langle L, Z\rangle$ has exactly the asserted maximum bias and variance and attains the quantile bound. Continuity, including the zero-variance limit, gives a minimum. Finally $F(h;\mu,v) \leq \Phi((h-\mu)/v)$ implies $Q_\alpha(\mu,v) \geq \mu + z_{1-\alpha}v$. If $a_0 \leq z_{1-\alpha}\tau$, every $t \in [0,1]$ requires radius at least a_0, attained at $t = 0$. □

At cap one, $M = \tau\sqrt{p}$ and $\alpha = 0.05$, the exact affine length is 2τ. The scalar bounded-normal-mean mechanism is classical [Stark, 1992]; the tensor reduction is proved above.

C.3 Noise envelopes and approximate ranks

Define $\Theta^\eta = \{A : \exists B \in \Theta(M, \bar{r}),\ \|A - B\|_F \leq \eta\}$ for supplied $\eta \geq 0$.

Proposition 12 (Envelope and approximation calibration)**.** *Suppose $d_i = a_i + e_i$ and, uniformly over the asserted class, $\mathbb{P}(\max_i |e_i| \leq \lambda_0) \geq 1 - \alpha$. For any measurable $\Lambda \geq \lambda_0$, the threshold center has an honest radius*

$$\|w\|_\infty D M^{2/K} (2\Lambda)^{1-2/K} + \eta$$

on Θ^η. The structural radius is $h_0 + \eta$. Data-dependent selection between these two intervals is valid. A direct Gaussian alternative needs its own calibration and deterministic selection or joint error allocation.

Roadmap. Use one event for every allowed threshold and a residual Lipschitz bound.

Proof. On the envelope event, Lemma 3 holds simultaneously for all $\Lambda \geq \lambda_0$, proving the assertion at $\eta = 0$. For fixed e, the residual map $f_e(x) = \mathrm{soft}_\Lambda(x + e) - x = e - \mathrm{clip}(x + e, [-\Lambda, \Lambda])$ is one-Lipschitz. For $a_i = \langle A, V_i\rangle$ and $b_i = \langle B, V_i\rangle$,

$$|\sum_i w_i\{f_{e_i}(a_i) - f_{e_i}(b_i)\}| \leq \|w\|_2 \|a - b\|_2 \leq \|A - B\|_F.$$

The last step is Bessel. Add the threshold error at B. Also $|\psi(A) - \psi(B)| \leq \|L\|_F \|A - B\|_F \leq \eta$. Both intervals cover on the same event, proving the selection assertion. □

For iid Gaussian errors the exact threshold is $\tau\Phi^{-1}(\{1 + (1-\alpha)^{1/m}\}/2)$, by multiplying the m probabilities $2\Phi(\lambda_0/\tau) - 1$. Dependent Gaussian margins with variances at most τ^2 admit $\tau\Phi^{-1}(1 - \alpha/(2m))$ by Bonferroni. Marginal bounds $\mathbb{E}e^{te_i} \leq e^{t^2v^2/2}$ give $v\sqrt{2\log(2m/\alpha)}$ by Chernoff and a union bound. Bounds $\mathbb{E}|e_i|^a \leq v^a$ give $v(m/\alpha)^{1/a}$ by Markov and a union bound. Independence is unnecessary for these three union-bound calibrations.

Writing $\mathcal{D}_L A = (\langle A, V_i\rangle)_i$, the same proof gives

$$|\widehat{\psi} - \psi(A)| \leq \inf_B \left\{ \|w\|_\infty D_B \|B\|_F^{2/K} (2\Lambda)^{1-2/K} + \|\mathcal{D}_L(A - B)\|_2 \right\}.$$

The infimum ranges over tensors with their indicated valid budget D_B. Its unknown value is not an adaptive confidence radius.

For Tucker approximation let P_k be any leading mode singular-space projector of dimension $\min(\bar{r}_k, p_k)$ and put $B = (\bigotimes_k P_k)A$. The lifted projections commute. The identity $I - \prod_k P_k = \sum_k (\prod_{\ell<k} P_\ell)(I - P_k)$ has summands with mutually orthogonal ranges. Their squared residual norms add, each at most $\|(I - P_k)A\|_F^2$. Thus

$$\|A - B\|_F^2 \leq \sum_k \sum_{j > \bar{r}_k} \sigma_j(A_{(k)})^2, \qquad \|B\|_F \leq \|A\|_F.$$

This holds without a spectral gap, including ties. The distance-class proposition also covers TT approximation.

C.4 Finite variance and bounded row corruption

Proposition 13 (Replicate median calibration)**.** *Let clean tensor replicates be independent with common mean A, and every prescribed contraction have variance at most known v^2. Allow at most c adversarial replacements of entire replicates. Coordinates within a replicate may be dependent. Choose an odd block count b with*

$$b \geq 8c, \qquad b \geq 32\log(m/\alpha), \qquad n \geq 2b, \qquad k = \lfloor n/b \rfloor.$$

Partition the first bk replicates into fixed disjoint blocks. For coordinatewise medians d_i of the block means,

$$\mathbb{P}(\max_i |d_i - a_i| > 2v/\sqrt{k}) \leq \alpha.$$

Thus Proposition 12 applies. For the smallest admissible odd b, the threshold has order

$$O\{v\sqrt{c/n + \log(m/\alpha)/n}\}.$$

Roadmap. Bound clean-block errors, count corrupted blocks deterministically and concentrate the original bad-block count.

Proof. Chebyshev bounds a clean block's error probability at $2v/\sqrt{k}$ by $1/4$. For a fixed coordinate let I_j be its original clean-block error indicators. They are independent and $\sum_j \mathbb{E}I_j \leq b/4$. At most c summaries change even for an adaptive adversary. If the final median errs, $\sum_j I_j \geq b/2 - c \geq 3b/8$. For a Bernoulli I, $h(t) = \log \mathbb{E}e^{t(I-\mathbb{E}I)}$ satisfies $h(0) = h'(0) = 0$ and $h''(t) \leq 1/4$, the variance of a tilted Bernoulli. Hence $h(t) \leq t^2/8$. Independence and Markov give

$$\mathbb{P}\{\sum_j (I_j - \mathbb{E}I_j) \geq b/8\} \leq e^{-tb/8 + bt^2/8} = e^{-b/32} \quad (t = 1/2).$$

A union bound over m coordinates proves the claim. Since $k \geq n/(2b)$, the rate follows, including the fixed rounding constants in b. □

Median-of-means calibration is classical [Devroye et al., 2016]. For independent Bernoulli replacements of probability ϵ, let $c = \lceil 2n\epsilon + 2\log(1/\delta) \rceil$. Their count satisfies $\mathbb{E}2^N = (1+\epsilon)^n \leq e^{n\epsilon}$, giving $\mathbb{P}(N > c) \leq e^{n\epsilon - c\log 2} \leq \delta$. With envelope error budget α_0, coverage is at least $1 - \alpha_0 - \delta$ when the block conditions hold. These sufficient corruption conditions are not a sharp breakdown or contamination-minimax result.

C.5 Unknown Gaussian variance

Proposition 14 (Independent residual calibration)**.** *Retain $n \geq 2$ iid Gaussian full-tensor replicates with unknown noise standard deviation $\sigma > 0$. Let $d_{ji} = \langle W_j, V_i \rangle$, $\bar{d}_i = n^{-1}\sum_j d_{ji}$, $\nu = m(n-1)$, and*

$$S^2 = \nu^{-1} \sum_i \sum_j (d_{ji} - \bar{d}_i)^2, \qquad \sigma_U = S\sqrt{\nu/\chi^2_{\nu,\delta}}, \quad 0 < \delta < \alpha.$$

The denominator is the lower δ chi-square quantile. Compute the original three intervals with $\tau_U = \sigma_U/\sqrt{n}$ and $\alpha_0 = \alpha - \delta$, choosing the smallest using only S and supplied quantities. The selected interval is honest at level $1 - \alpha$. For fixed δ and $\nu \to \infty$, $\sigma_U/\sigma \to 1$ in probability, and the nonlinear expected-length order is preserved.

Roadmap. Orthogonally separate means and residuals, then condition on a conservative scale estimate.

Proof. An orthogonal transformation in the replicate index separates each sample mean from its $n-1$ residual coordinates. Joint Gaussianity and loading orthonormality give independence of all residual and mean coordinates. Therefore $\nu S^2/\sigma^2 \sim \chi^2_\nu$, independent of the vector of means, and $\mathbb{P}(\sigma_U \geq \sigma) = 1-\delta$. Conditional on S with $\sigma_U \geq \sigma$, both stochastic intervals have failure at most α_0 and the selected index is fixed. The structural interval covers surely. Integrating conditional failure and adding δ proves honesty.

The normalized chi-square has mean one and variance $2/\nu$. Chebyshev gives convergence to one and convergence of its fixed-δ quantile to one. Thus $\sigma_U/\sigma \to 1$ and $\mathbb{E}(\sigma_U/\sigma)^2 = \nu/\chi^2_{\nu,\delta} \to 1$. The second-moment bound gives uniform integrability for powers at most one. Since the nonlinear radius is proportional to $\tau_U^{1-2/K}$, its expected order is preserved. □

Retained residuals are essential to this construction; it is not an estimate of variance from Z alone.

C.6 Confidence levels and covariance

Proposition 15 (Calibrated lower bounds)**.** *The fixed-confidence rates of Theorems 6, 4 and Proposition 1 extend to every $0 < \alpha < 1/2$ after changing constants. If known Gaussian covariance satisfies $\Sigma \succeq s_0^2 I$ with $s_0 > 0$, all-procedure isotropic lower bounds transfer with $\tau = s_0$.*

Roadmap. Tune the mixture distance below $1-2\alpha$; compare covariance models by adding independent noise.

Proof. Fix $0 < \eta < 1-2\alpha$. For the coordinate mixture use $a = \min\{M, \tau\sqrt{\log(1+4\eta^2 p)}\}$, giving $\chi^2 \leq 4\eta^2$. For the unfolding mixture use $\theta = \min(M, c_\eta \tau\sqrt{p})$ with $c_\eta^2 = \{1-(1+4\eta^2)^{-2}\}/2$. Its second-moment bound is at most $1+4\eta^2$. For the diagonal support mixture use

$$b_s = \min\{Ms^{-K/2},\ \tau\sqrt{\log[1+\log(1+4\eta^2)p/s^2]}\}.$$

The factorial-moment bound again gives second moment at most $1+4\eta^2$. Total variation is at most η in each case; the interval lemma has positive factor $1-2\alpha-\eta$. The fixed-α rates are unchanged.

Add independent $N(0, \Sigma - s_0^2 I)$ noise to $N(A, s_0^2 I)$. Applying any Σ-procedure to the result gives an isotropic randomized procedure with the same coverage and length distributions. The isotropic all-procedure lower bound therefore applies. □

Upper envelope calibration uses the marginal variances $\mathrm{vec}(V_i)^\top \Sigma\, \mathrm{vec}(V_i)$; the direct radius uses $z_{1-\alpha/2}\sqrt{\mathrm{vec}(L)^\top \Sigma\, \mathrm{vec}(L)}$. A lower covariance bound is needed for a universal positive noise lower scale: if $\Sigma = \tau^2(I - \mathrm{vec}(L)\,\mathrm{vec}(L)^\top)$, the target is observed without noise. Cross terms $B^\top \Sigma^{-1} C$ cannot in general be ordered by a covariance bound; the randomization proof avoids that invalid step.

D Proofs of the sharp rates and computational statements

D.1 Proof of Theorem 9

Roadmap. Embed a harmonic rank-one profile on a random support; bound its full tensor likelihood using support overlap.

Proof. For a size-m support S, assign values $j^{-1/3}$, $j = 1, \dots, m$, in increasing coordinate order and zeros elsewhere; call this vector x_S. Set $A_S = b_m x_S^{\otimes 3}$. Then

$$\|A_S\|_F = b_m \left(\sum_{j=1}^{m} j^{-2/3} \right)^{3/2} \le b_m \sqrt{27m} \le M, \qquad \psi(A_S) = b_m H_m / \sqrt{p}.$$

The sum bound follows by integrating $x^{-2/3}$. All alternatives have rank one and the same target. For $H = |S \cap T|$, Cauchy–Schwarz and the decreasing profile give

$$0 \le \langle x_S, x_T \rangle \le \left(\sum_{i \in S \cap T} x_{S,i}^2 \sum_{i \in S \cap T} x_{T,i}^2 \right)^{1/2} \le \sum_{j=1}^{H} j^{-2/3} \le 3H^{1/3}.$$

At $H = 0$ every term is zero. Therefore $\langle A_S, A_T \rangle \le 27 b_m^2 H$. The full Gaussian likelihood identity and the hypergeometric bound already proved in Proposition 1 imply

$$1 + \chi^2(Q, P_0) \le \mathbb{E} e^{27 b_m^2 H / \tau^2} \le \exp\{(m^2/p)(e^{27 b_m^2/\tau^2} - 1)\} \le e^{1/4}.$$

Only the hypergeometric inequality is reused, not the diagonal likelihood. Lemma 5 proves (32).

For $p \ge 256$, $m = \lfloor p^{1/4} \rfloor \ge p^{1/4}/2$, so $H_m \ge \log m \ge (\log p)/8$. Also $p/(4m^2) \ge \sqrt{p}/4 \ge p^{1/4}$, and

$$\tfrac{1}{4} \log p \le \log(1 + p/(4m^2)) \le \log p \le \sqrt{p} \le p/m.$$

For $p \ge 4$, $\log p \le \sqrt{p}$ follows from positivity at four and the nonnegative derivative of $\sqrt{p} - \log p$. Thus the noise term is the minimum in b_m and is at least $\tau \sqrt{\log p}/(2\sqrt{27})$. Multiplication proves the stated constant. □

D.2 Proof of Theorem 10

Roadmap. Verify measurable extrema and coverage, then bound differences of feasible tensors using doubled caps.

Proof. The parameter class is compact: its norm ball is bounded and closed, and rank constraints are closed polynomial conditions on unfolding minors. Each finite-net maximum is continuous. Hence every nonempty feasible section is compact and its functional extrema are attained.

The feasible-data domain is closed. Indeed, if $Z_j \to Z$ and B_j are feasible, compactness supplies a subsequence $B_{j_k} \to B \in \Theta$; continuity preserves every constraint. For any real u, the set of Z admitting a feasible B with $\psi(B) \ge u$ is also closed by this argument, as is the analogous set with $\psi(B) \le u$. The extrema on the feasible-data domain, extended by zero elsewhere, are therefore Borel functions. The noise event at error level α makes A feasible, uniformly with probability at least $1 - \alpha$. On that event C_F covers the target.

For $B, B' \in \mathcal{F}(Z)$, put $\Delta = B - B'$. Matrix-rank subadditivity gives the doubled unfolding caps; coordinate restriction cannot increase them. Also $N_S(\Delta) \le N_S(Z - B) + N_S(Z - B') \le 2\lambda_s$, so $\rho(\Delta_S) \le 4\lambda_s$. The diagonal bound and injective-norm comparison with J_2 give

$$\sum_{i \in S} |\Delta_{i,\dots,i}|^{2/K} \le J_2^{1-1/K} \|\Delta_S\|_F^{2/K} \le J_2 \rho(\Delta_S)^{2/K}.$$

Taking the s largest diagonal entries yields $|\Delta|_{(s)} \le 4 J_2^{K/2} \lambda_s s^{-K/2}$. Summation proves $|\psi(B) - \psi(B')| \le W_\alpha$. Apply it to the attained extrema; the structural range gives the other bound $2h_0$. Empty sections have length zero.

For fixed confidence, $\lambda_s \le C_{K,\alpha}\tau\sqrt{s\log p}$ for large p. The sum of $s^{-(K-1)/2}$ is bounded for $K \ge 4$ and at most $1+\log p$ for $K=3$. This proves the worst-case upper bounds. Theorem 6 matches the higher-order bound and Theorem 9 matches the cubic bound, already at zero and within the rank-one subclass. Since $\mathcal{L}_0 \le \mathcal{L}_{\max}$, both rates follow. □

D.3 Proof of Proposition 2

Roadmap. Use a smaller failure budget and bounded target range for the upper risk; reduce estimation to common-target testing for the lower risk.

Proof. For $p \ge 2$, $\alpha_p = p^{-2}$ still gives $W_{\alpha_p} = O(r_K(p))$. The midpoint is in $[-h_0, h_0]$, including the empty-set convention. On coverage its error is at most $W_{\alpha_p}/2$; otherwise the error is at most $2h_0$. Thus

$$\sup_A \mathbb{E}_A(\widehat{\psi} - \psi(A))^2 \le W_{\alpha_p}^2/4 + 4h_0^2 p^{-2} = O(r_K(p)^2),$$

since $h_0 = \tau\sqrt{s_{\mathrm{cap}}}$. For a common-target prior with target δ, test whether an estimator is at least $\delta/2$. If its worst-case squared risk is R, Markov bounds the total null-plus-mixture testing error by $8R/\delta^2$. Total variation bounds this error below by $1 - \mathrm{TV}(P_0, Q)$. Hence $R \ge \delta^2(1-\mathrm{TV})/8$. The coordinate and sparse harmonic mixtures have bounded total variation and targets of the required respective orders. □

D.4 Proof of Proposition 3

Roadmap. Combine leading-entry scan bounds with a quasi-norm tail bound and an explicit grid-net construction.

Proof. On acceptance and the common noise event, $|a|_{(s)} \le 4J^{K/2}\lambda_s s^{-K/2}$ for $s \le t$. The remaining magnitudes are at most a_t, so

$$\sum_{s>t} |a|_{(s)} \le a_t^{1-q} \sum_{s>t} |a|_{(s)}^q \le a_t^{1-q} B_q.$$

The tail vanishes for $t = p$. This proves the radius. On rejection the structural interval covers surely. Intersection gives honesty by error allocation and length at most ℓ_*. At zero rejection has probability at most γ, giving the stated expectation bound.

For the rate, substitute $\lambda_s \lesssim \tau\sqrt{s\log p}$ and $B_q = J^{1-1/K}\tau^{2/K}p^{1/K}$. The tail becomes a constant times $\tau p^{1/K-1/2}(\log p)^{(1-2/K)/2} t^{-(K-1)(1-2/K)/2}$, as asserted.

For a computable net, split each coordinate of $[-1,1]^s$ into $N = \lceil 4K\sqrt{s}\rceil$ equal intervals and normalize every nonzero product-grid vector. Rounding a unit vector incurs error at most $\sqrt{s}/N \le 1/(4K)$. The rounded vector is nonzero, and normalization adds at most the original error, giving a $1/(2K)$-net. Its size is at most $(N+1)^s \le (3+4K\sqrt{s})^s$. Use log cardinality $Ks\log(3+4K\sqrt{s})$ in the Gaussian union bound. Since $s \le p$, this still gives $\lambda_s \lesssim_K \tau\sqrt{s\log p}$. Each contraction costs $O(s^K)$ operations, proving the complexity bound. All completed levels cover on one event; adaptive continuation and intersection add no scan error. The separately calibrated C_* is intersected only once. □

D.5 Proof of Theorem 11

Roadmap. Expand in Hermite polynomials, bound overlap moments and control the possible mean shift of a low-degree estimator by its null variance.

Proof. Scale by τ and let h_ν be normalized multivariate probabilists' Hermite polynomials over the p^K coordinates. The generating identity $e^{tz-t^2/2} = \sum_{j\geq 0} H_j(z)t^j/j!$ gives $\mathbb{E}H_j(g)H_k(g) = j!\mathbf{1}_{\{j=k\}}$: multiply two generating functions and compare coefficients of their expectation e^{tu}. Coordinate independence gives multivariate orthonormality. The shift calculation similarly gives $\mathbb{E}h_\nu(g+a) = a^\nu/\sqrt{\nu!}$. The coefficient identities follow by differentiation at zero of Gaussian entire moment-generating functions.

For the finite support prior, the degree-d coefficient sum is

$$\sum_{|\nu|=d} \{\mathbb{E}_Q h_\nu(Z/\tau)\}^2 = \frac{1}{d!}\mathbb{E}_{S,T}\left(\frac{\langle A_S, A_T\rangle}{\tau^2}\right)^d = \frac{\lambda^{2d}}{d!s^{Kd}}\mathbb{E}H^{Kd},$$

by the multinomial theorem, where $H = |S \cap T|$. This uses the full likelihood.

Let $\mathrm{S}(m,j)$ count partitions of an m-element set into j nonempty unlabeled sets. Counting maps according to their number of image elements gives $H^m = \sum_{j=1}^m \mathrm{S}(m,j)(H)_j$. There are at most j^m labeled maps, so $\mathrm{S}(m,j) \leq j^m/j!$. For $j \leq s$, the hypergeometric factorial moments obey

$$\mathbb{E}(H)_j = (s)_j^2/(p)_j \leq (s^2/p)^j = \mu^j,$$

using $(s-\ell)/(p-\ell) \leq s/p$ and $s - \ell \leq s$ factor by factor. For $j > s$ the moment is zero. Therefore

$$\mathbb{E}H^m \leq \sum_{j=1}^m j^m \mu^j/j! \leq m^m(e^\mu - 1) \leq \mu e^\mu m^m.$$

The centered projection omits degree zero. Summing the remaining degrees,

$$\chi^2_{\leq D} \leq \mu e^\mu \sum_{d=1}^D \frac{\lambda^{2d}(Kd)^{Kd}}{d!s^{Kd}} \leq \mu e^\mu \sum_{d=1}^D B_D^d \leq \mu e^\mu B_D/(1-B_D) = \eta_D^2.$$

For a degree-D estimator with worst-case squared risk R on zero and the alternatives, $|\mathbb{E}_0 T| \leq \sqrt{R}$ and $|\mathbb{E}_Q T - \delta| \leq \sqrt{R}$. Orthonormal projection and Cauchy–Schwarz give

$$|\mathbb{E}_Q T - \mathbb{E}_0 T| \leq \sqrt{\mathrm{Var}_0\, T}\sqrt{\chi^2_{\leq D}} \leq \sqrt{R}\,\eta_D.$$

Thus $\delta \leq (2+\eta_D)\sqrt{R}$, proving the risk lower bound. For the stated exponents, $\mu = O(p^{2a-1}) \to 0$ and $B_D = O(p^{1-aK+bK}) \to 0$; floor factors remain bounded for large p. The target is $\delta = \tau s^{1-K/2}$. Since $a(K-2) < 1$, its square is polynomially larger than τ^2/p times either logarithmic factor of the unrestricted minimax risk. The alternatives have norm M and rank one in every unfolding, so belong to either positive cap class. □

D.6 Proof of Proposition 4

Roadmap. Combine mixture coverage with null coverage and Markov's inequality for length.

Proof. Mixture coverage gives $Q(\delta \notin C) \leq \alpha$. If $0 \in C$ and $|C| < \delta$, the interval cannot contain δ. Consequently

$$P_0(\delta \in C) \leq P_0(0 \notin C) + P_0(|C| \geq \delta) \leq \alpha + \mathbb{E}_0|C|/\delta.$$

Add the errors. Polynomial-time interval evaluation and membership evaluation yield a polynomial-time test, proving the conditional implication. No unrestricted testing-hardness conclusion is deduced from the low-degree squared-risk bound. □

E Proofs for norm budgets and loading geometry

E.1 Proof of Theorem 12

Roadmap. Bound each ordered diagonal difference twice, then take the minimum before summing.

Proof. The coverage, compactness, and measurable-extrema argument of Theorem 10 applies for every finite $M > 0$. For two feasible tensors put $\Delta = B - B'$. Then $\|\Delta\|_F \le 2M$ and every constrained unfolding rank is at most twice its original cap. Writing $a_i = \Delta_{i,\dots,i}$ and $q = 2/K$, Theorem 1 gives

$$s|a|_{(s)}^q \le \sum_i |a_i|^q \le J_2^{1-1/K}(2M)^q.$$

The norm argument and the residual-scan proof give, respectively,

$$|a|_{(s)} \le P_2 M s^{-K/2}, \qquad |a|_{(s)} \le Q_2 \lambda_s(\alpha) s^{-K/2}.$$

Both inequalities hold for the same ordered entry. Summing their minimum bounds $|\psi(B) - \psi(B')|$ by (37). The structural range gives $2h_0$. Empty feasible sets produce a singleton and therefore length zero. □

Lemma 8 (A norm-sensitive harmonic sum)**.** *For $p \ge 1$, $x > 0$ and $K \ge 3$, put*

$$S_{K,p}(x) = \sum_{s=1}^{p} \min\{x s^{-K/2}, s^{-(K-1)/2}\}.$$

Then $S_{3,p}(x) \le 4g_p(x)$, and $S_{K,p}(x) \le 3\min(x, 1)$ for $K \ge 4$. For fixed K and $0 < \alpha < 1$, $U_{K,p}(M;\alpha) \le C_{K,\bar r,\alpha}\mathfrak{r}_K(p, M, \tau)$ uniformly in p, M, τ.

Roadmap. Split the cubic sum at $s = x^2$ and use summability in higher order.

Proof. For $K \ge 3$, $S_{K,p}(x) \le x\sum_{s\ge1} s^{-3/2} \le 3x$. For $K \ge 4$ the second term gives $S_{K,p}(x) \le \sum_{s\ge1} s^{-3/2} \le 3$.

For $K = 3$, $S_{3,p}(x) \le H_p \le \ell_p$. If $x \le 1$, the bound $3x$ suffices for $g_p(x) = x$. If $1 < x < \sqrt{p}$, put $n = \lfloor x^2 \rfloor$. Since $n \ge x^2/2$,

$$S_{3,p}(x) \le H_n + x\sum_{s>n} s^{-3/2} \le 1 + 2\log x + 2x/\sqrt{n} \le 4(1 + \log x).$$

If $x \ge \sqrt{p}$, the bound $H_p \le 1 + \log p \le 2(1 + \log x)$ gives the same conclusion. Together with $3x$ and ℓ_p, this proves the first statement.

The threshold satisfies

$$\lambda_s(\alpha) \le c_{K,\alpha}\tau\sqrt{s\ell_p}, \qquad c_{K,\alpha} = \{2[2 + K\log(1 + 4K) + \log(2/\alpha)]\}^{1/2}.$$

Indeed, bound $\log(ep/s) \le \ell_p$, $\log p \le \ell_p$, $s \ge 1$, and $\ell_p \ge 1$ in (25). The inequality $\min(au, bv) \le \max(a, b)\min(u, v)$ for nonnegative arguments shows

$$U_{K,p}(M;\alpha) \le \max(P_2, Q_2 c_{K,\alpha})\frac{\tau\sqrt{\ell_p}}{\sqrt{p}} S_{K,p}(x).$$

The established sum bounds prove the claim. □

E.2 Proof of Theorem 13

Roadmap. Combine the new diameter with a coordinate mixture and a norm-tuned harmonic support mixture.

Proof. The upper bound follows from Theorem 12 and Lemma 8. Since the upper diameter is deterministic it controls both expected-length criteria.

For the lower bound put $a = \min\{M, \tau\sqrt{\log(1+p/4)}\}$ and mix uniformly over aE_i, $i = 1, \dots, p$. These rank-one alternatives have target $a/\sqrt{p}$. The full Gaussian likelihood identity gives

$$\chi^2(Q, P_0) = \frac{e^{a^2/\tau^2} - 1}{p} \le \tfrac{1}{4}, \qquad \mathrm{TV}(Q, P_0) \le \tfrac{1}{4}.$$

The common-target interval lemma implies

$$\mathbb{E}_0|C| \ge (3/4 - 2\alpha)\frac{\min\{M, \tau\sqrt{\log(1+p/4)}\}}{\sqrt{p}}. \tag{51}$$

For $p \ge 1$, $\log(1+p/4) \ge \ell_p/5$. To verify this, the difference is positive at one and its derivative is $1/(p+4) - 1/(5p) \ge 0$. Thus (51) is at least a constant times $\tau\sqrt{\ell_p}\min(x,1)/\sqrt{p}$. This proves the lower bound for every $K \ge 4$, and for $K = 3$ when $x \le 1$.

For the remaining cubic case assume first $p \ge 256$ and $x > 1$, and take

$$u = \min(p^{1/4}, x^2), \qquad m = \lfloor u \rfloor \ge 1.$$

Use the sparse harmonic mixture of Theorem 9, with the original general-budget coefficient b_m. Since $m \le x^2$, $M/\sqrt{m} \ge \tau\sqrt{\ell_p}$. Also $m \le p^{1/4}$ implies

$$\log(1 + p/(4m^2)) \ge \tfrac{1}{4}\log p \ge \ell_p/5,$$

where the first inequality holds for $p \ge 256$ and the second for $\log p \ge 4$. Hence $b_m \ge \tau\sqrt{\ell_p}/\sqrt{135}$. Furthermore $H_m \ge 1$ and $H_m \ge \log(m+1) \ge \log u$. Writing $t = \log x$, if $2t \le (\log p)/4$ then $H_m \ge \max(1, 2t) \ge (1+t)/2$. Otherwise $H_m \ge (\log p)/4 \ge \ell_p/5$. In both cases

$$H_m \ge \tfrac{1}{5}\min(1 + \log x, \ell_p) = \tfrac{1}{5} g_p(x),$$

where $1 + \log x \le x$ for $x \ge 1$ was used. Consequently

$$\mathbb{E}_0|C| \ge \frac{c_{\alpha,d}}{5\sqrt{135}}\mathfrak{r}_3(p, M, \tau). \tag{52}$$

All likelihoods here are for the full tensor; the support-mixture proof holds for the newly chosen m and b_m without modification.

For $p < 256$, $\ell_p < 7$ and $g_p(x) \le 7\min(x, 1)$, so the coordinate mixture suffices with a changed absolute constant. Finally $\mathcal{L}_0 \le \mathcal{L}_{\max}$ completes the proof. Every lower alternative has rank one; higher supplied caps enter only the upper constants. □

Lemma 9 (Residual fit and its noise tail)**.** *The estimator in (40) is Borel measurable. With $T_0 = T(\tau G)$ and $\beta = 1/8$,*

$$\mathbb{P}(T_0 > t) \le \beta e^{-t^2/(2\tau^2)} \quad (t \ge 0), \qquad \mathbb{E}T_0^2 \le 2\beta\tau^2.$$

Put $z_K = \sum_{s=1}^{\infty} s^{-K/2} \le 3$. Then

$$|\widehat{\psi} - \psi(A)| \le U_{K,p}(M; \beta) + \frac{Q_2 z_K}{\sqrt{p}}(T_0 + \tau/2), \qquad |\widehat{\psi} - \psi(A)| \le 2h_0. \tag{53}$$

Roadmap. Use a Gaussian union bound with a common additive slack, then compare an approximate residual minimizer to the truth.

Proof. Each unit net contraction is one-Lipschitz in the Frobenius norm; finite maxima and the positive part preserve this property. Thus T is continuous and one-Lipschitz. Density of (B_j) implies $d(Z) = \min_{B\in\Theta} T(Z - B)$, although the estimator only uses a positive tolerance. The function d is one-Lipschitz as an infimum of functions with the same Lipschitz constant. The sets $\{Z : T(Z - B_j) \le d(Z) + \tau\}$ are Borel, and their union is the entire observation space. Selecting the first such index is measurable. Hence so is $\widehat{\psi}$.

For $t \ge 0$, apply the same finite Gaussian union bound as for the original scan at thresholds $\lambda_s(\beta) + t$. Since

$$e^{-(\lambda_s+t)^2/(2\tau^2)} \le e^{-\lambda_s^2/(2\tau^2)} e^{-t^2/(2\tau^2)},$$

the total probability is at most $\beta e^{-t^2/(2\tau^2)}$. Integrating the tail of T_0^2 proves its moment bound.

For $\widehat{A} = B_{j(Z)}$, comparison with A gives $T(Z - \widehat{A}) \le d(Z) + \tau \le T_0 + \tau$. Therefore for every support,

$$N_S(\widehat{A} - A) \le 2\lambda_s(\beta) + 2T_0 + \tau.$$

The net comparison and doubled-rank argument bound its ordered diagonal entry by $Q_2(\lambda_s(\beta) + T_0 + \tau/2)s^{-K/2}$. It is also bounded by $P_2 M s^{-K/2}$. Use $\min(a, b + c) \le \min(a, b) + c$, sum, and divide by $\sqrt{p}$ to obtain (53). Both tensors belong to the original cap class, so their targets lie in $[-h_0, h_0]$, giving the second bound. □

E.3 Proof of Theorem 14

Roadmap. Control residual-fit tails without an M-dependent failure event and reuse the same full-data mixtures for the lower bound.

Proof. Lemma 9 and $(a + b)^2 \le 2a^2 + 2b^2$ give

$$\sup_A \mathbb{E}_A(\widehat{\psi} - \psi(A))^2 \le \min\left\{4h_0^2, 2U_{K,p}(M;\beta)^2 + (8\beta + 1)Q_2^2 z_K^2 \frac{\tau^2}{p}\right\}. \tag{54}$$

If $x \le 1$, use $h_0^2 \le s_{\mathrm{cap}} M^2/p$ and $\mathfrak{r}_K = M/\sqrt{p}$. If $x > 1$, then $\mathfrak{r}_K \ge \tau\sqrt{\ell_p}/\sqrt{p} \ge \tau/\sqrt{p}$; Lemma 8 absorbs both terms in the second bound into a constant times $\mathfrak{r}_K^2$.

The lower constructions in Theorem 13 have common target $\delta \ge c\mathfrak{r}_K$ and total variation bounded by a constant below one, independently of p, M, τ. For any estimator with worst-case risk R, the test $\mathbf{1}\{\widetilde{\psi} \ge \delta/2\}$ has total null-plus-mixture error at most $8R/\delta^2$ by Markov's inequality. Testing error is at least $1 - \mathrm{TV}(P_0, Q)$. Thus $R \ge \delta^2(1 - \mathrm{TV})/8 \gtrsim \mathfrak{r}_K^2$. □

E.4 Proof of Theorem 15

Roadmap. Rearrange weights against ordered diagonal bounds; equalize the target of a hard mixture by a contraction in one mode.

Proof. Extend each prescribed orthonormal loading family to an orthonormal mode basis. Mode transformations preserve Gaussian isotropy and every constrained unfolding rank; restriction to the loading block cannot increase rank or Frobenius norm. Apply residual projection to that m^K block.

Every pair of feasible block tensors satisfies the ordered-entry bound from Theorem 12, with $p = m$. The rearrangement inequality gives

$$|\psi(B) - \psi(B')| \le \sum_{s=1}^{m} w_s^* |\operatorname{diag}(B - B')|_{(s)} \le U_w.$$

The structural bound and the earlier measurable-extrema argument give the other assertions about this interval.

For the lower bound fix $d \le m$ and work in the coordinates corresponding to the d largest absolute weights. These weights are nonzero. Take whichever coordinate or nonnegative cubic harmonic mixture supplies the lower rate in dimension d in Theorem 13. Its common normalized diagonal target is at least $c\mathfrak{r}_K(d, M, \tau)$. In the first tensor mode apply the diagonal linear map with multipliers w_d^*/w_i on these coordinates. Its operator norm is at most one, so every alternative remains rank one with Frobenius norm at most M. Its weighted target is now the same for every alternative and equals $w_d^*\sqrt{d}$ times the original normalized target.

For each pair of alternatives, the first-mode factor inner product becomes

$$\sum_i (w_d^*/w_i)^2 u_i v_i \le \sum_i u_i v_i,$$

because the original factors are nonnegative. The other factor inner products are unchanged and nonnegative. Thus the full tensor inner product does not increase. The Gaussian likelihood second moment is no larger than that of the original hard mixture. Embedding these alternatives in the full ambient tensor adds only independent mean-zero noise coordinates and leaves the likelihood second moment unchanged. The common-target interval lemma therefore gives $\mathbb{E}_0|C| \ge c_{K,\alpha} w_d^* \sqrt{d}\, \mathfrak{r}_K(d, M, \tau)$. Maximize over the finite set of d to obtain (42). □

E.5 Proof of Proposition 5

Roadmap. Rotate to the loading block, absorb known signs, and apply the matching dimension-m theorem.

Proof. Replace each first-mode loading vector by $\epsilon_i v_{1i}$, preserving orthonormality. The loading now has equal positive weights $m^{-1/2}$. The reduced m^K Gaussian tensor has the same norm and cap constraints, so Theorems 13 and 14 give the upper bounds. For the lower bounds embed their rank-one alternatives in the loading block and set all other mean entries to zero. The extra independent noise carries no parameter information; the full likelihood second moments equal the reduced ones. Equivalently, the interval lower bound follows by taking $d = m$ in Theorem 15, and the same mixtures give the squared-risk lower bound. This includes $m = 1$ and arbitrary ambient dimensions at least m. □

F Proofs for robustness, loading transfer, and adaptation

F.1 Proof of Theorem 16

Roadmap. Identify the tail inequalities actually used by the upper proofs, then use a Gaussian subfamily for family-wise minimax lower bounds.

Proof. For every unit net direction, Chernoff applied to (43) gives $\mathbb{P}(|\langle U, E\rangle| > t) \le 2e^{-t^2/(2v^2)}$. The finite union bound proving scan coverage and the additive-slack bound proving Lemma 9 use no

independence between directions. Both therefore hold with v replacing τ. All subsequent diameter and squared-error inequalities are deterministic functions of these events and moments. Orthogonal changes of coordinates preserve (43). The weighted proofs apply unchanged. The unknown-law family includes $E \sim N(0, v^2 I)$. Honesty over the family implies honesty over that subexperiment, and the worst-case risk or expected length over the family is at least its Gaussian value. The Gaussian lower bounds give the matching order. □

F.2 Proof of Theorem 17

Roadmap. Control every median by a binomial tail, obtain a common excess-tail moment, and reuse the ordered-difference argument.

Proof. For a direction at level s, each block average has variance at most σ^2/k_s. Chebyshev gives a block-error probability at most $\sigma^2/(k_s t^2)$. If the median errs by more than t, at least half its blocks err. Independence of blocks, followed by a union over their subsets, gives

$$\mathbb{P}(|d_U - \langle U, A\rangle| > \lambda_s^{\mathrm{mom}} + t) \le 2^{-b_s}(1 + t/\lambda_s^{\mathrm{mom}})^{-b_s}, \quad t \ge 0.$$

This follows by substituting $\lambda_s^{\mathrm{mom}} = 4\sigma/\sqrt{k_s}$ into $2^{b_s}\{\sigma^2/[k_s(\lambda_s^{\mathrm{mom}} + t)^2]\}^{b_s/2}$. There are at most e^{H_s} directions at level s. By the definition of b_s, $\sum_s e^{H_s} 2^{-b_s} \le \beta$. Thus all constraints cover the true tensor with probability at least $1 - \beta$.

Since $k_s \ge n/(2b_s)$, $\lambda_s^{\mathrm{mom}} \le 4\sqrt{2}\tau\sqrt{b_s}$. Define $T_0 = \max_U(|d_U - \langle U, A\rangle| - \lambda_{s(U)}^{\mathrm{mom}})_+$. For $a \ge 0$, the function $b\log(1 + a/\sqrt{b})$ is increasing for $b > 0$: its derivative is $\log(1 + u) - u/[2(1 + u)] \ge 0$, $u = a/\sqrt{b}$. Because every $b_s \ge 9$, the union bound yields

$$\mathbb{P}(T_0 > t) \le \beta(1 + t/(12\sqrt{2}\tau))^{-9}, \qquad \mathbb{E}T_0^2 \le \frac{72}{7}\beta\tau^2. \tag{55}$$

For the moment, integrate $2t$ times the tail and use $\int_0^\infty u(1 + u)^{-9}du = 1/(8 \cdot 7)$.

For two feasible tensors every directional difference is at most $2\lambda_s^{\mathrm{mom}}$. The net comparison, doubled caps and norm bound prove exactly the previous norm-sensitive diameter. Medians are measurable finite order statistics. The extrema of their compact feasible sections are Borel by the same compact-section argument as before, now treating $(d_U)_U$ as the finite observation vector. The smallest-odd-integer rule gives $b_s \le 11 + [H_s + \log(p/\beta)]/\log 2 \le C_{K,\beta} s\ell_p$. The sum lemma therefore bounds the diameter by $C\mathfrak{r}_K$.

For estimation, F is continuous and one-Lipschitz in B. Selecting the first dense-sequence element with value at most $\inf_j F(B_j) + \tau$ is measurable. Its comparison with A gives directional differences at most $2\lambda_s^{\mathrm{mom}} + 2T_0 + \tau$. Lemma 9's deterministic argument gives

$$|\widehat{\psi} - \psi(A)| \le U^{\mathrm{mom}} + Q_2 z_K(T_0 + \tau/2)/\sqrt{p}, \qquad |\widehat{\psi} - \psi(A)| \le 2h_0.$$

Equation (55) and the small-M structural bound prove risk $O(\mathfrak{r}_K^2)$ exactly as in Theorem 14. For the lower bound restrict the replicate family to iid Gaussian errors with covariance $\sigma^2 I$. Orthogonal transformation in the replicate index separates the sample mean, whose noise level is $\sigma/\sqrt{n}$, from mean-independent residuals. Thus the full replicate likelihood second moments equal the sample-mean ones. The earlier Gaussian lower bounds apply to any procedure using all replicates, including randomized ones. Finally, H_s is increasing in s, so the replicate condition has the stated sufficient order p for fixed K, β. □

F.3 Proof of Proposition 6

Roadmap. Exhibit exact scalar recovery and a correlated direction with inflated variance.

Proof. If $|A| < v$, then $\operatorname{sign}(A + v\xi) = \xi$. The estimator $Z - v\operatorname{sign}(Z)$ is identically A, proving (i). The Rademacher moment-generating function obeys $\cosh(tv) \le e^{t^2v^2/2}$. For (ii), take the unit product tensor with every entry $p^{-K/2}$. Its contraction with E is $v\xi p^{K/2}$, with variance v^2p^K. Each individual entry still has variance and sub-Gaussian proxy v^2. □

F.4 Proof of Theorem 18

Roadmap. Inflate scan constraints for a bounded approximation residual, then combine two nested lower-bound subclasses.

Proof. Write $A = B_0 + R$, $B_0 \in \Theta$, $\|R\|_F \le \eta$; compactness ensures such a decomposition exists. On the Gaussian scan event every residual contraction of $Z - B_0$ is at most $\lambda_s + \eta$, so B_0 is feasible. Inflate its projected target interval by η on both sides. If the feasible set is empty use a singleton; this cannot occur on the coverage event. Since $\|L\|_F = 1$, the enlarged interval covers $\psi(A)$. For feasible differences the diameter argument gives

$$|C| \le U_{K,p}(M;\alpha) + \{Q_2 z_K/\sqrt{p} + 2\}\eta,$$

using $\min(a, b + c) \le \min(a, b) + c$. It also gives $|C| \le 2h_0 + 2\eta$. Choose between this procedure and the direct Gaussian interval using their deterministic length bounds. This proves the upper order $\min\{\tau, \mathfrak{r}_K + \eta\}$.

The class contains Θ, giving lower bound $c\mathfrak{r}_K$ at zero. It also contains $\{tL : |t| \le \eta\}$ because $0 \in \Theta$. Comparing zero with tL for $t = \min(\eta, \tau/2)$ gives total variation at most $\frac{1}{2}\sqrt{e^{1/4} - 1} < 1/2$ and target difference t. Hence the lower bound is also $c_\alpha \min(\eta, \tau)$. Finally $\min(\tau, r + \eta) \le r + \min(\tau, \eta) \le 2\max\{r, \min(\tau, \eta)\}$ for every $r \ge 0$. This establishes both criteria up to fixed constants. □

F.5 Proof of Proposition 7

Roadmap. Embed sparse diagonal tensors of rank s, then compare with the full norm ball.

Proof. For a uniform size-s support S, set $A_S = b\sum_{i\in S} E_i$ with

$$b = \min\{M/\sqrt{s}, \tau\sqrt{\log(1 + p/(4s^2))}\}.$$

All constrained ranks are s, its norm is $b\sqrt{s} \le M$, and its target is $sb/\sqrt{p}$. The full tensor inner product is $b^2|S \cap T|$; the hypergeometric factorial-moment inequality gives $1 + \chi^2(Q, P_0) \le \exp\{(s^2/p)(e^{b^2/\tau^2} - 1)\} \le e^{1/4}$. The common-target lemma proves the lower bound for every $s \le r_*$. Maximize over s. If there are no rank restrictions, the line $\{tL : |t| \le M\}$ belongs to the Frobenius ball. The two-point argument with $t = \min(M, \tau/2)$ gives lower order $\min(M, \tau)$. The direct interval and $[-M, M]$, chosen deterministically by radius, give the matching upper bound for every mean in that ball. No asymptotic fixed-cap argument applies to this enlarged class. □

F.6 Proof of Proposition 8

Roadmap. Add an independently calibrated error event, without assuming stochastic independence.

Proof. On the intersection of $\langle L_0, A\rangle \in C_0$ and $|\langle R, E\rangle| \le q_R$, adding the second target contribution gives the asserted interval. A union bound controls failure by $\alpha_0 + \alpha_1$. The length is the displayed sum. Chernoff or the Gaussian distribution of $\langle R, E\rangle$ supplies the calibrations. For a prescribed partition, repeat the argument over blocks using the sum of their allocated failures; their centers may be dependent. Selecting among complete constructions from deterministic diameter bounds uses no data and retains its own coverage guarantee. Applying the result with $L_0 = 0$ also covers every L, although it then reduces to the direct interval. □

F.7 Proof of Theorem 19

Roadmap. Match elementary constructive upper bounds with the existing small-budget lower bound and a product-direction two-point test.

Proof. In (i), the rate formula has $x \le 1$ and is $M/\sqrt{p}$. The structural radius is $M\sqrt{s_{\mathrm{cap}}/p}$, with fixed cap factor. Theorem 13 gives the matching lower bound. For (ii), choose the sign of U so that $\langle L, U\rangle \ge c\|L\|_F$ and compare zero with tU, $t = \min(M, \tau/2)$. The full likelihood second moment is $e^{t^2/\tau^2} \le e^{1/4}$ and the target difference is at least $ct\|L\|_F$. The common-target lemma gives the lower order at zero. Cauchy–Schwarz supplies structural radius $M\|L\|_F$, while the direct Gaussian radius is $z_{1-\alpha/2}\tau\|L\|_F$. Their deterministic selection controls worst-case length and is polynomial in the number of observed coordinates. The cap class is used only for the lower inclusion; fixed rank caps are unnecessary here. □

F.8 Proof of Proposition 9

Roadmap. Find a one-dimensional Gaussian submodel inside every such convex relaxation.

Proof. The average of ME_i, $i = 1, \dots, p$, is $(M/\sqrt{p})L$. Both signs and zero belong to the convex hull, hence so does the asserted line segment. Compare zero with tL for $t = \min(M/\sqrt{p}, \tau/2)$ and apply the same two-point calculation. At $M = \tau\sqrt{p}$ it forces order τ. The previous minimax rates divided by τ tend to zero for each fixed $K \ge 3$. This argument changes the honesty class and makes no complexity assumption. □

F.9 Proof of Theorem 20

Roadmap. Shift the full-data hard mixtures into an unused orthogonal block.

Proof. Embed the dimension-m rank-one alternatives from Theorem 13, with norm budget M_0, in coordinates $\{q+1, \dots, p\}^K$. Denote them by B. Their supports are disjoint from A_0, so $\|A_0 + B\|_F^2 = \|A_0\|_F^2 + \|B\|_F^2 \le M^2$. Unfolding rank subadditivity shows $A_0 + B$ is in the outer cap class. Translation of Gaussian likelihoods by A_0 leaves their second moments unchanged. All alternatives have a common increment in the normalized p-diagonal target equal to $\sqrt{m/p}$ times their dimension-m target. Their total variation from the law at A_0 is bounded by a fixed constant less than $1 - 2\alpha$. Apply the common-target lemma with baseline target $\psi(A_0)$, or translate the interval by that fixed number. This proves the lower bound. □

F.10 Proof of Theorem 21

Roadmap. Recover ranks on one spectral event, bound a second-order projection residual, and condition on the independent pilot.

Proof. For each unfolding use 1/4-nets of its two unit spheres, of total cardinality at most $9^{p_k+d_k}$. Product telescoping bounds its operator norm by twice the maximum net contraction. Chernoff and a union bound over the K unfoldings give an event $\mathcal{E}$ of probability at least $1-\delta$ on which every pilot error operator norm is at most η_k. No independence between unfoldings is used. The singular-value perturbation inequality $|\sigma_j(Z_{(k)}^{(1)}) - \sigma_j(A_{(k)})| \le \eta_k$ follows from the variational characterization of singular values: for every unit vector the two image norms differ by at most η_k, and taking the relevant min/max over subspaces preserves this bound. Thus zero singular values produce observed values at most η_k, while nonzero values exceeding $2\eta_k$ produce observed values greater than η_k. All selected ranks are exact on $\mathcal{E}$. If any is zero then $A = 0$ and the singleton covers.

For a positive selected rank r_k, write $A_{(k)} = U_k S_k V_k^\top$ using its true nonzero singular values. Because Q_k is the leading observed rank-r_k projector,

$$\|(I - Q_k)A_{(k)}\|_{\mathrm{op}} \le \sigma_{r_k+1}(Z_{(k)}^{(1)}) + \eta_k \le 2\eta_k.$$

For full row rank the residual is zero, and the same inequality holds. Right multiplication by $V_k S_k^{-1}$ gives

$$\|(I - Q_k)U_k\|_{\mathrm{op}} \le \frac{2\eta_k}{\sigma_{r_k}(A_{(k)})} \le \frac{2\eta_k}{\sigma_{r_k}(Z_{(k)}^{(1)}) - \eta_k}.$$

Clipping at one yields d_k^*. This calculation uses the gap only on the stated separated class, with a strictly positive observed denominator.

Expand the identity into the 2^K mutually orthogonal products of Q_k and $I - Q_k$. The projector T retains exactly the terms with at most one complement. Each term discarded by T appears in at least one product $(I - Q_k)(I - Q_\ell)$. Orthogonality therefore implies

$$\|(I - T)A\|_F^2 \le \sum_{k<\ell} \|(I - Q_k)(I - Q_\ell)A\|_F^2 \le M^2 \sum_{k<\ell} (d_k^* d_\ell^*)^2.$$

The last inequality uses $A = (\bigotimes_k U_k)G$ with $\|G\|_F = \|A\|_F$ and the operator norm of a product of mode maps. Also $\|(I-T)A\|_F \le M$. Since T is orthogonal, Cauchy–Schwarz bounds $|\langle L, (I-T)A\rangle|$ by B_L.

Condition on the pilot. The vector TL is now deterministic, and independence makes the second error contraction sub-Gaussian with proxy $v_2\|TL\|_F$. Conditional on any pilot in $\mathcal{E}$, the projected interval has error at most $\alpha - \delta$. The direct alternative has the same conditional failure bound and the structural alternative never fails. A choice determined by the pilot thus also has conditional failure at most $\alpha - \delta$. Intersecting with a surely valid structural interval preserves coverage; empty intersections may be replaced by a singleton. Integrating and adding $\mathbb{P}(\mathcal{E}^c) \le \delta$ proves honesty. SVDs, finite sums, and tensor matrix products give the stated arithmetic complexity.

For Gaussian artificial splitting the two centered errors are jointly Gaussian with zero cross covariance, each of covariance $2\tau^2 I$, hence independent. Zero covariance would not suffice for non-Gaussian errors. No minimax length conclusion on the original unseparated class follows from this proof. □

G Result status and remaining questions

PROVED denotes a complete supplied argument, with finite checks where useful. It does not denote independent peer review, formal proof verification or certified priority.

Item	Scope	Status
Theorems 1–6	Original geometry, honest interval and comparison bounds	PROVED
Proposition 1	Classical diagonal-information optimum	PROVED
Theorems 7–8	Initial full-data scan and harmonic obstruction	PROVED
Theorem 9	Sharp cubic sparse harmonic lower bound	PROVED
Theorem 10	Matching full-data zero-point and worst-case lengths	PROVED
Proposition 2	Full-data squared-error minimax rates	PROVED
Proposition 3	Certified finite-scan search tradeoff	PROVED
Theorem 11	Restricted polynomial-estimation obstruction	PROVED
Proposition 4	Interval-to-testing implication	PROVED
Appendix C	Calibrated geometric and noise refinements	PROVED
Theorems 12–13	Full-data length for every norm budget	PROVED
Theorem 14	Uniform norm-budget squared risk	PROVED
Theorem 15	Orthogonal weighted upper/lower envelopes	PROVED
Proposition 5	Sharp flat-support and rotated-loading rates	PROVED
Original Q2	Coordinate lower scale for all $K \geq 3$	REFUTED
Q3	Actual-rank adaptation and sharp general loading geometry	OPEN
Q4	Polynomial-time minimax confidence intervals	OPEN

Table 7: Updated ledger. The earlier cubic and worst-case statistical gaps are closed for fixed caps; the norm-budget restriction is removed by the new results. The low-degree result is not an unrestricted computational-hardness theorem.

The earlier ledger above describes the baseline weak-signal problem. Section 7 adds the following proved results with their exact scope:

- Theorems 16 and 17: full-data joint sub-Gaussian upper bounds and finite-variance replicate rates, with noise-family minimax statements and the required replication.
- Proposition 6: counterexamples to universal per-law lower bounds and marginal-only scan calibration.
- Theorem 18 and Proposition 7: sharp approximation-allowance length, a growing-rank lower bound, and the no-cap endpoint.
- Proposition 8 and Theorem 19: certified arbitrary-loading upper constructions and efficient minimax small-budget/product-overlap regimes.
- Proposition 9: an obstruction for honesty on the entire convex outer class, not general algorithmic hardness.
- Theorems 20 and 21: a nonzero-baseline lower bound and a computable rank-free interval on a separated mode-rank class.

Efficient minimax diffuse-loading weak-signal intervals, sharp all-loading/growing-cap dependence, and unseparated actual-rank adaptation remain open. The separated procedure is conservative at moderate signal strength. The new experiments evaluate implemented procedures and failure modes; the ideal minimax projection and robust residual optimizer have not been implemented. Unknown norm/noise bounds, arbitrary replicate dependence, and data-selected loadings remain outside the new claims. The real-data and LLM limitations persist. Independent proof and priority review is required before submission.

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